

**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Entrepreneurship Academy

ET5218 INNOVATION AND COMMERCIALIZATION OF TECHNOLOGIES

04 Technical Sketching

NTU Innovation and
Entrepreneurship

Developed by Kenny Hong



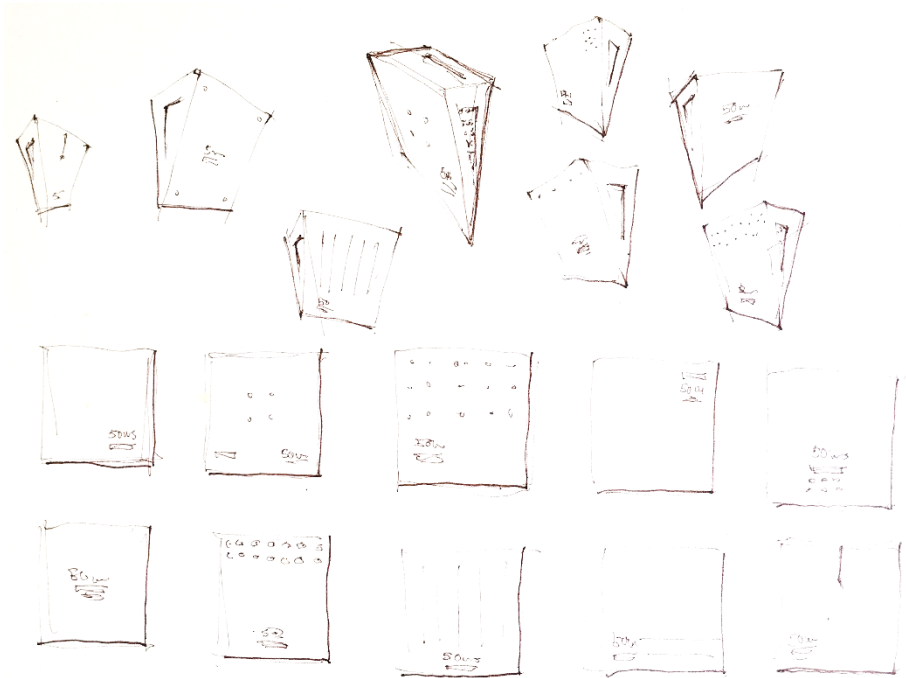
kenny hong

tp product design in 94'

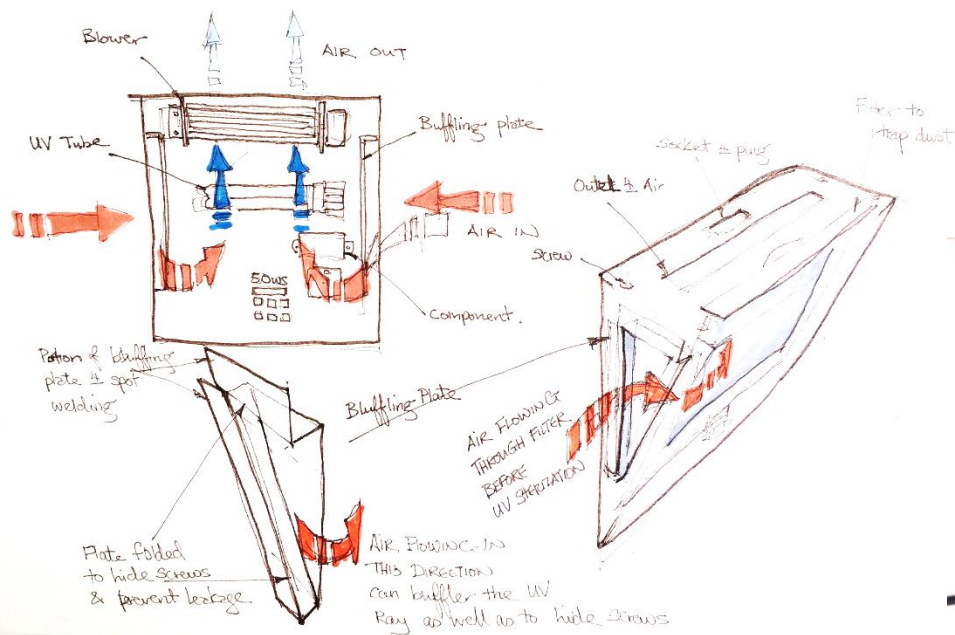
domus academy design in 07'

**head of product development design
studio 14'**

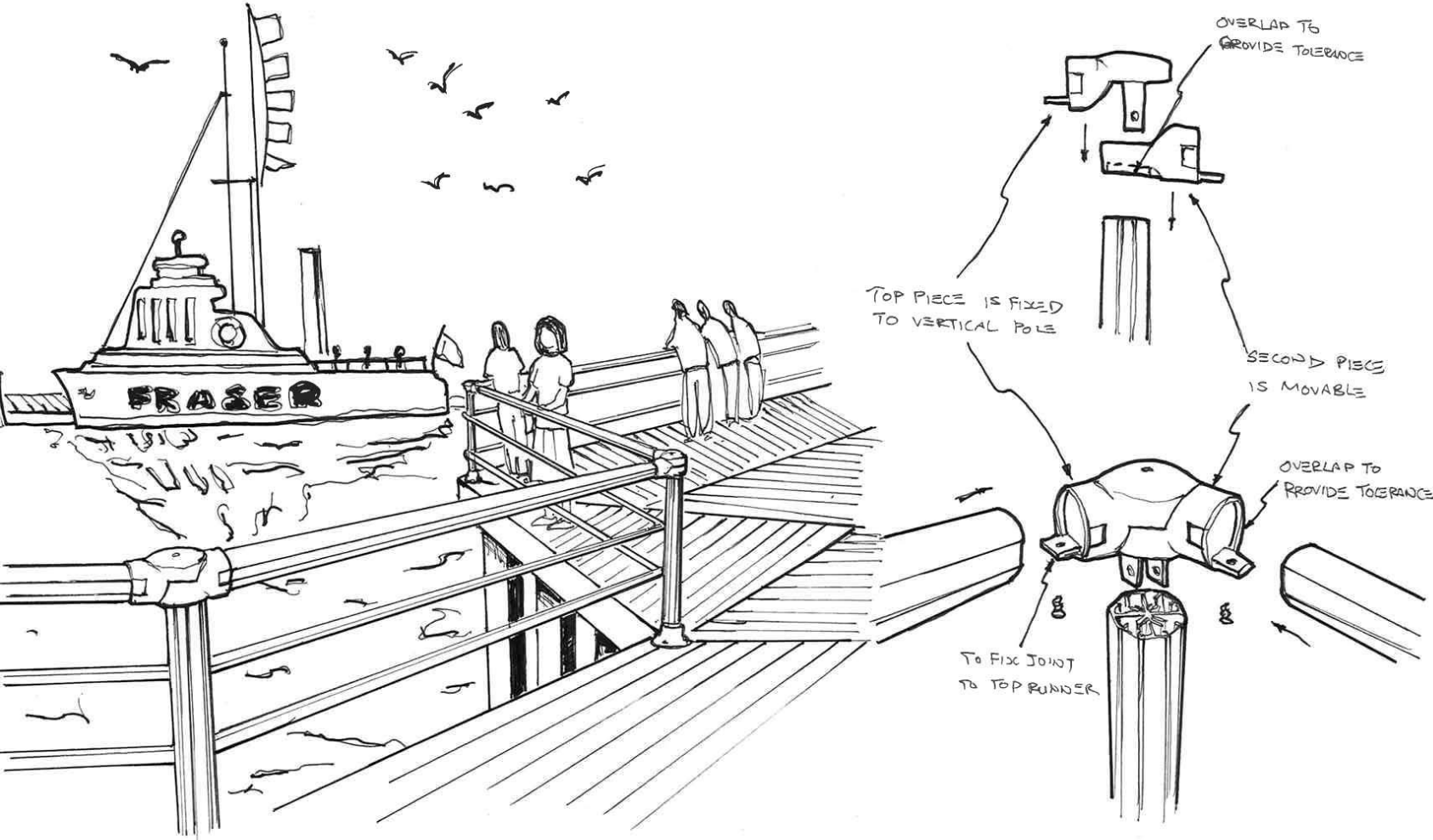
lectures at nafa & ntu



AIR FLOW AND BUFFLING PROBLEM.



ALUMINIUM STRUCTURES



TOP RUNNER & TOP RUNNER CLIP

The top runner is connected to the post via a Top Runner Clip, which is bolted down to the post through a pre-extruded slot

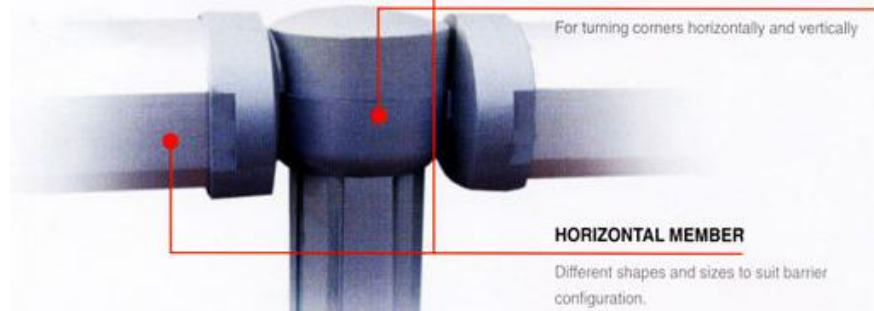


POST

The core component of QUICKLOCK™. There is a range of posts which cater for light, normal, and heavy duty pedestrian barriers. Each post has four recessed toothed slots for attaching horizontal members

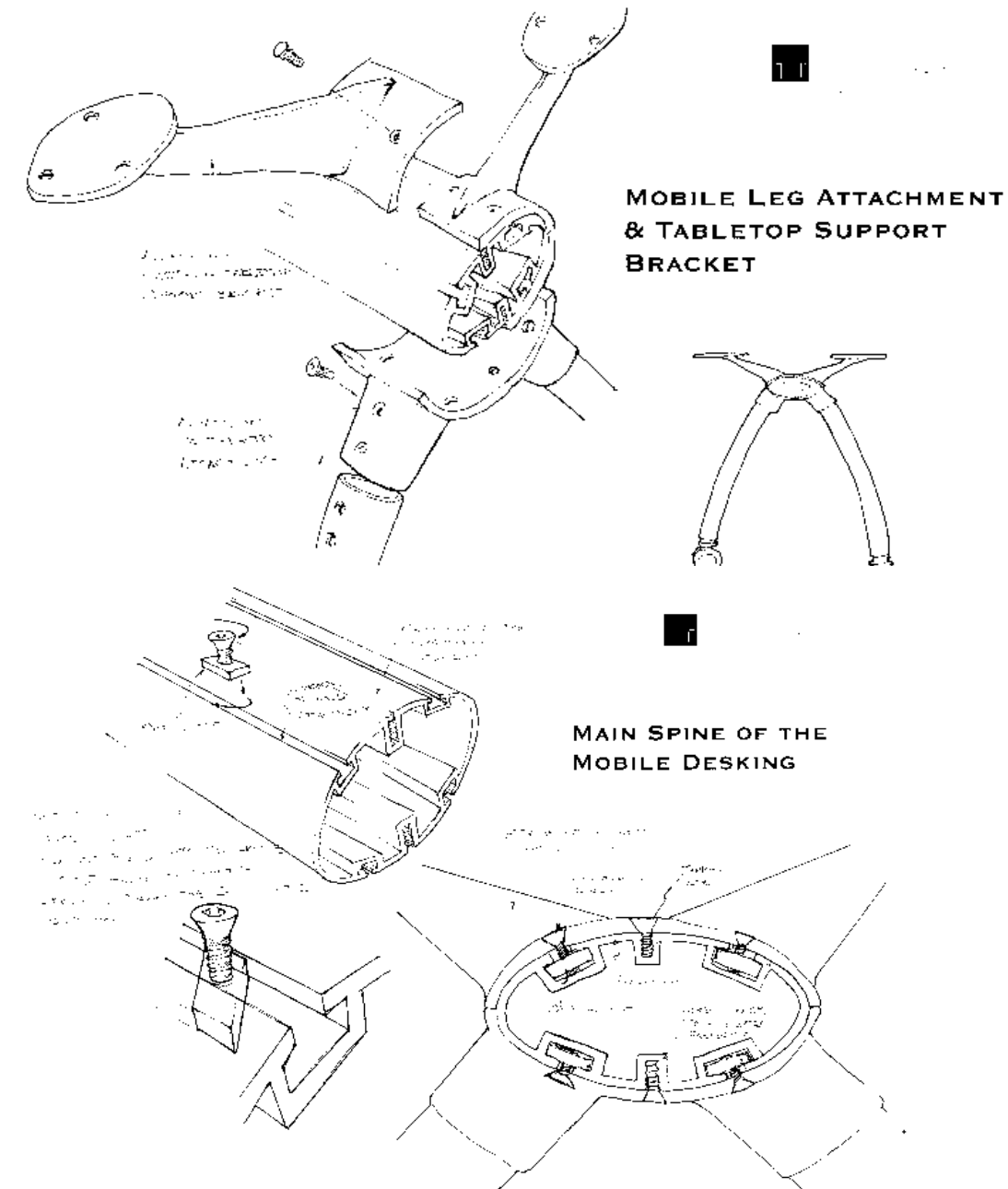
DISCONNECT™ TURNING MECHANISM

For turning corners horizontally and vertically



HORIZONTAL MEMBER

Different shapes and sizes to suit barrier configuration.







Decor that welcomes you home

Michelle Ng

When it comes to furnishing a home, most home-owners will focus on areas like the kitchen, dining area, master bedroom or even television console wall.

Rarely will they pay much attention to the entrance way. Interior and furniture designer Kenny Hong hopes to change that mindset.

"The entrance way, no matter the size, is the first space you see when you enter your home. It's where you set the tone for your house and an impression for your visitors," says the 51-year-old founder of local design studio 104.

"If people need a bench or a stool to put on their shoes, they'll either just go out and randomly buy one that fulfills that need, but not the design aesthetic," he adds.

To bring attention to this often underutilised space, Mr. Hong is launching a collection of small accessories and furniture that are typically found in the home's entrance at this year's GSS Experience Singapore.

Called Objects by the Entrance Way, the collection puts an aesthetic spin on nine commonly used products such as mirrors, rugs, umbrellas, stands and small stools.

Prices range from \$40 for a tissue box to \$825 for a mirror, which Mr Hong says is a slightly more "premium" price point as all the items are designed and manufactured in Singapore. Only the rugs (\$380) are produced in China.

All pieces will be displayed and sold at the Great Furniture Showcase at Robinsons. The event runs from June 21 to July 28.

The collection's simple design aesthetic involves the use of basic geometric shapes such as circles and triangles, along with clean straight

lines and curves to create interesting-looking products that are equally functional.

For example, an oblong mirror (\$825) is juxtaposed with a straight solid oak wood ledge attached to it, and a round wall clock (\$105) has a contrasting crescent face of both white and natural wood.

Mr Hong purposely kept the designs minimalist to not have the entrance overshadow the rest of the home.

"For each product, I thought about what is really needed and subtracted what was redundant to

keep it simple but still interesting," he says.

Mr Hong, who has 22 years of experience in the interior and product design field, finds satisfaction in taking commonly used items and elevating them to more interesting designs.

"You know that moment when you look at something familiar, but it's not quite the same because it's something you've never seen before? That's very refreshing and intriguing," he says.

It is a lesson he imparts to his students at his part-time lecturing

stint at Nanyang Academy of Fine Arts and Nanyang Technological University, where he teaches design fundamentals, furniture design and technical subjects.

But ultimately, his goal is to make good design more accessible to the public. He says: "Most people only think of fashion labels when they think of local designers, but there are so many more good locally designed products beyond clothing."

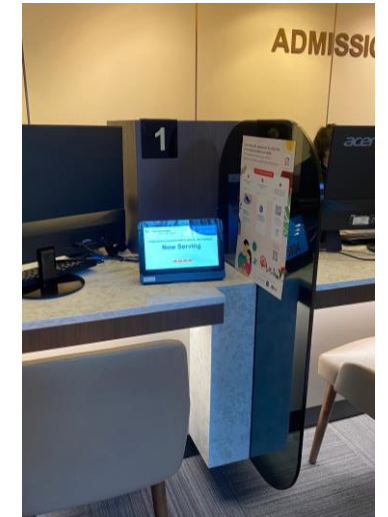
"To design a product that is useful to people and can be used in their homes is something I enjoy very much."

Design studio 104's Kenny Hong is launching a collection of accessories and furniture typically found in the home's entrance at this year's GSS Experience Singapore. ST PHOTO: ZHANG XIAO

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Object by the Entrance way











Learning Outcomes

1. Understand the fundamentals of product visual presentation.
2. Explain the Role of Visual Presentation in Entrepreneurial Pitching.

Technical Sketching

1. Introduction to Types of Drawing
2. Getting Ready with 5 Tips
3. Mastering the Pen
4. 5 Essentials to a Good Technical Sketching
5. Perspective Drawing
6. Equally Spaced Receding Elements
7. Drawing Cylindrical Part
8. Final Exercise

Drawing

a description of the object

it represents reality

an illusion

communicates both ideas and emotions



Types of Graphic Communication

sketches

drawings (technical sketching)

technical drawing

rendering

Sketches

first and quickest form of communication

visual thinking

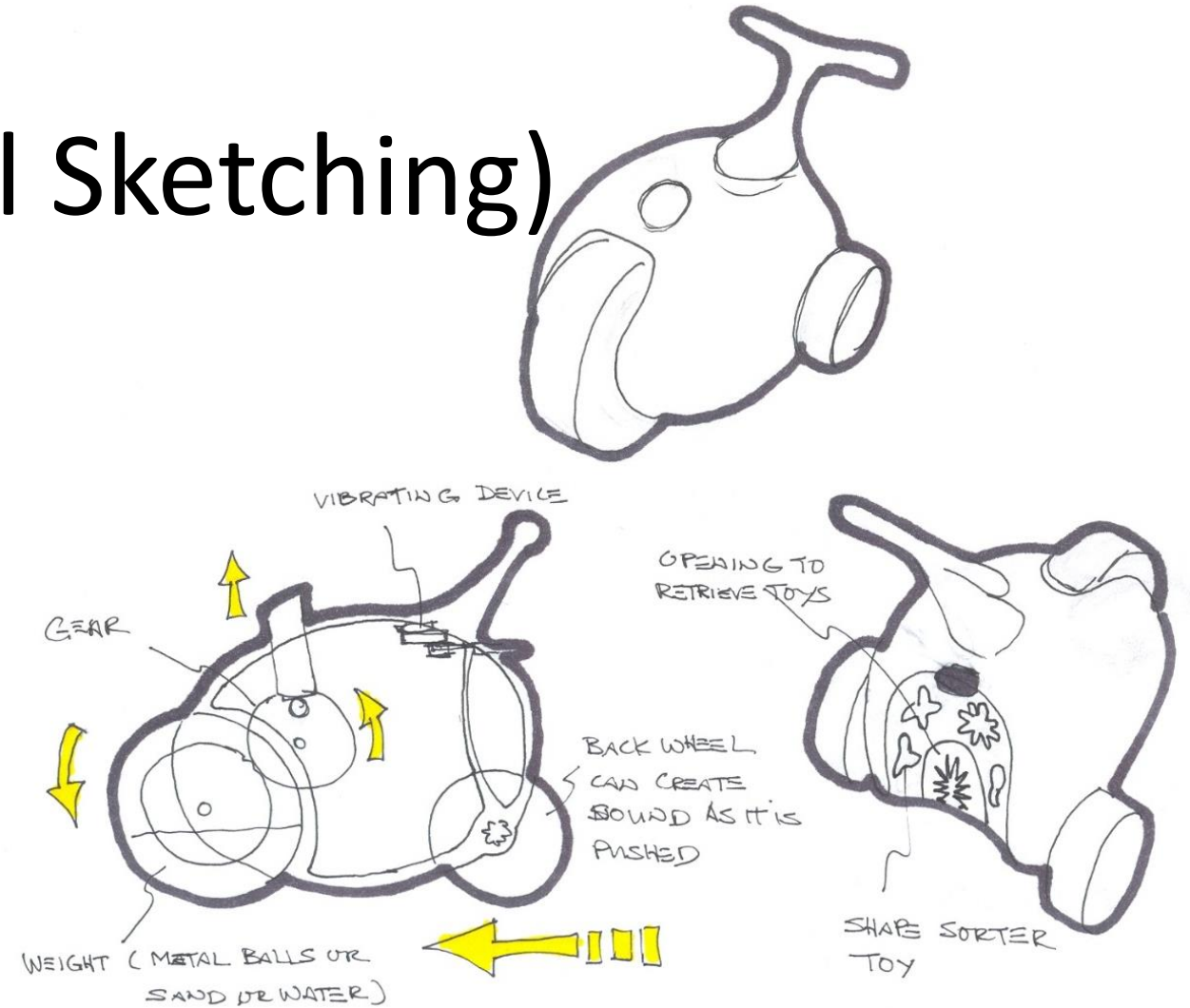


Drawings (Technical Sketching)

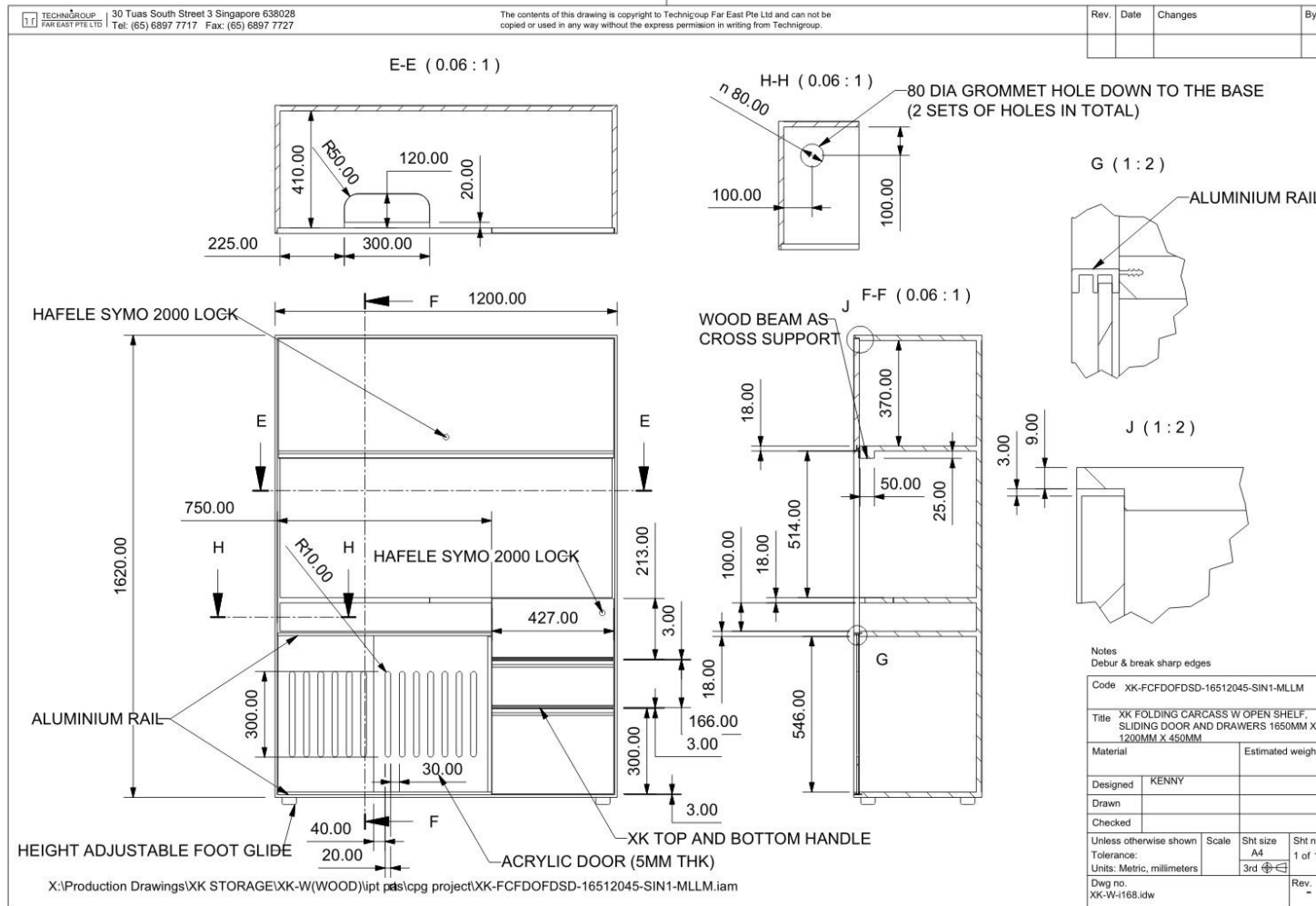
the stage after sketching

more finish image

easier to understand



Technical drawing



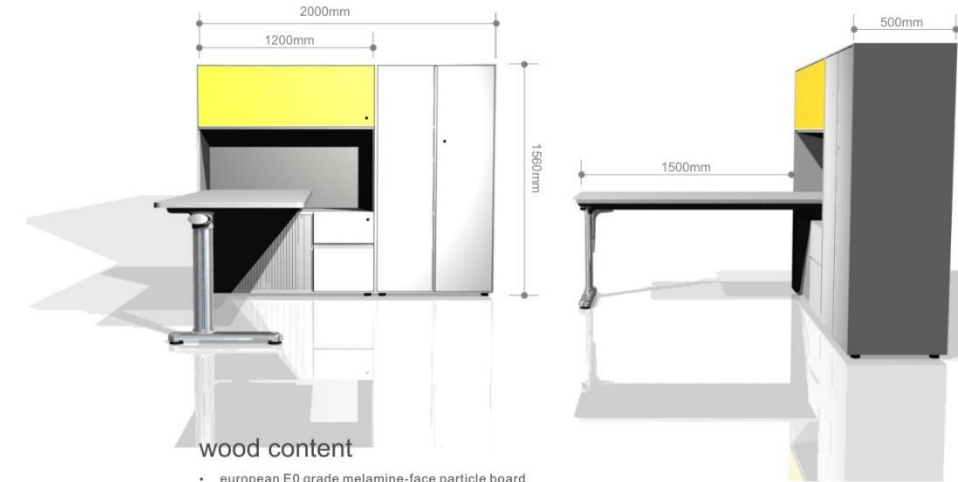
accurate, dimensioned and scaled drawings

drawn in special way, with special rules

Rendering

dramatized presentation of the design

sales tool



wood content

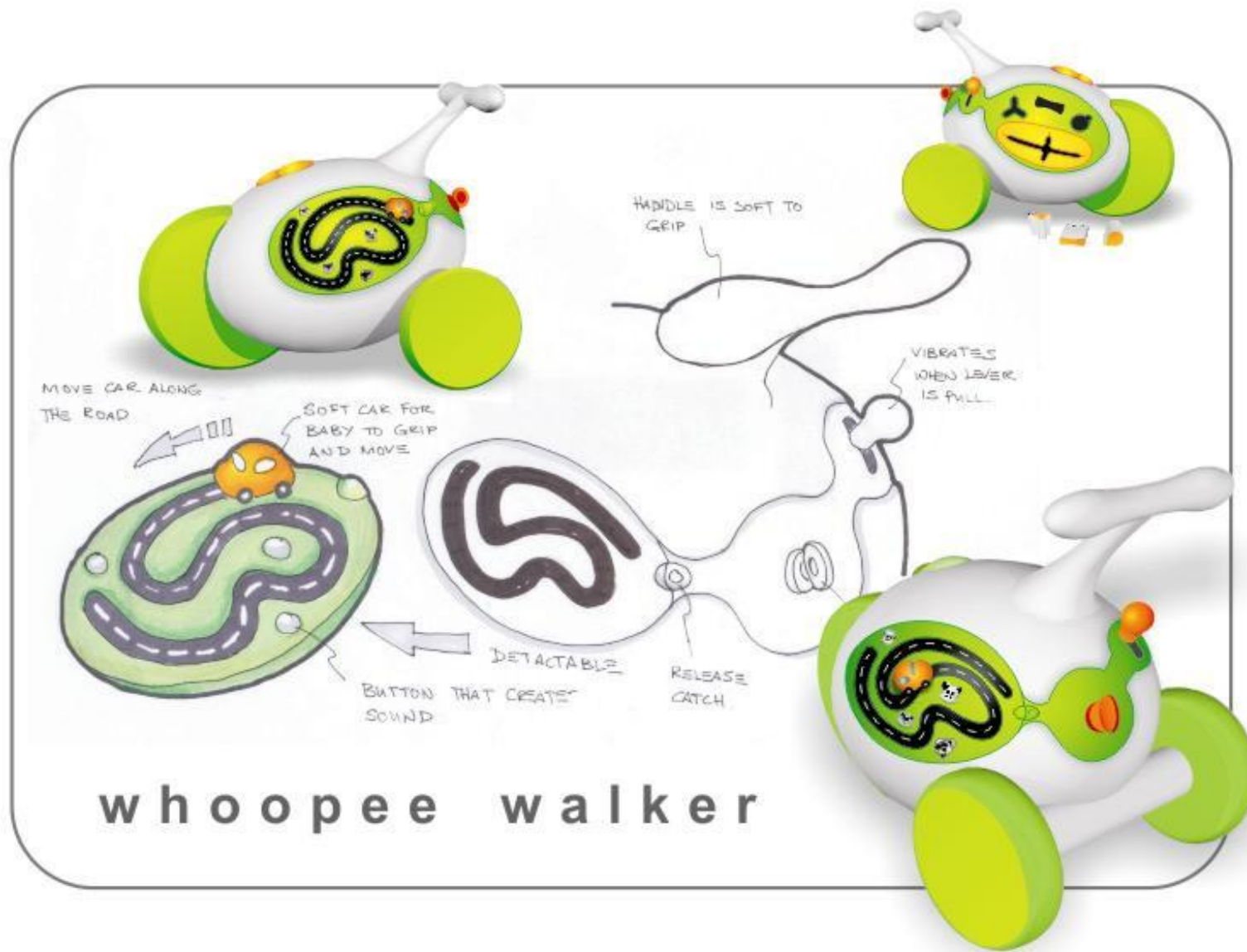
- european E0 grade melamine-face particle board
- up 70% pre-consumer content
- formaldehyde emission of less than 50 parts per billion (greenguard compatible)
- TVOC emission of less than 0.5mg/cubic meter (greenguard compatible)
- min density of 620kg/mc
- impact resistant 2mm PVC edging

aluminium handle

- aluminium extruded from grade 6063 alloy, strengthened with T5 treatment
- surfaces are treated with 10 microns-thick corrosion-resistant, anodising layer

X2 L leg desking

- high strength and lightweight aluminium alloy grade ADC-12 die casting components
- aluminium extruded from grade 6063 alloy, strengthened with T5 treatment
- satin polished finish
- aluminium extruded under table spine to prevent table from warping





"babe magnet"



get your baby to start developing fast with the whoopee walker

another development toy from

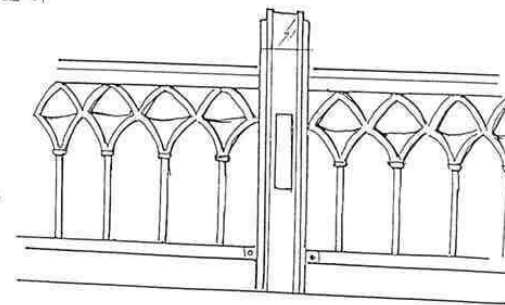
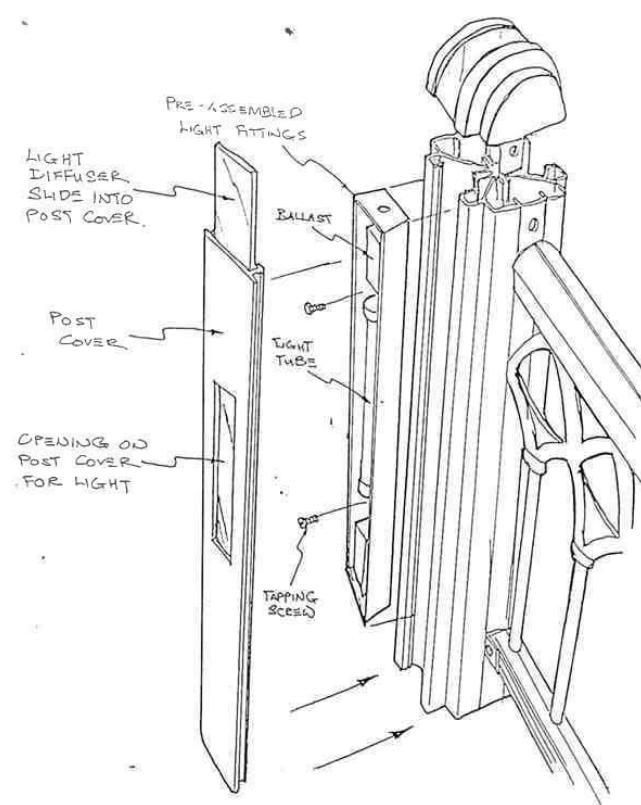
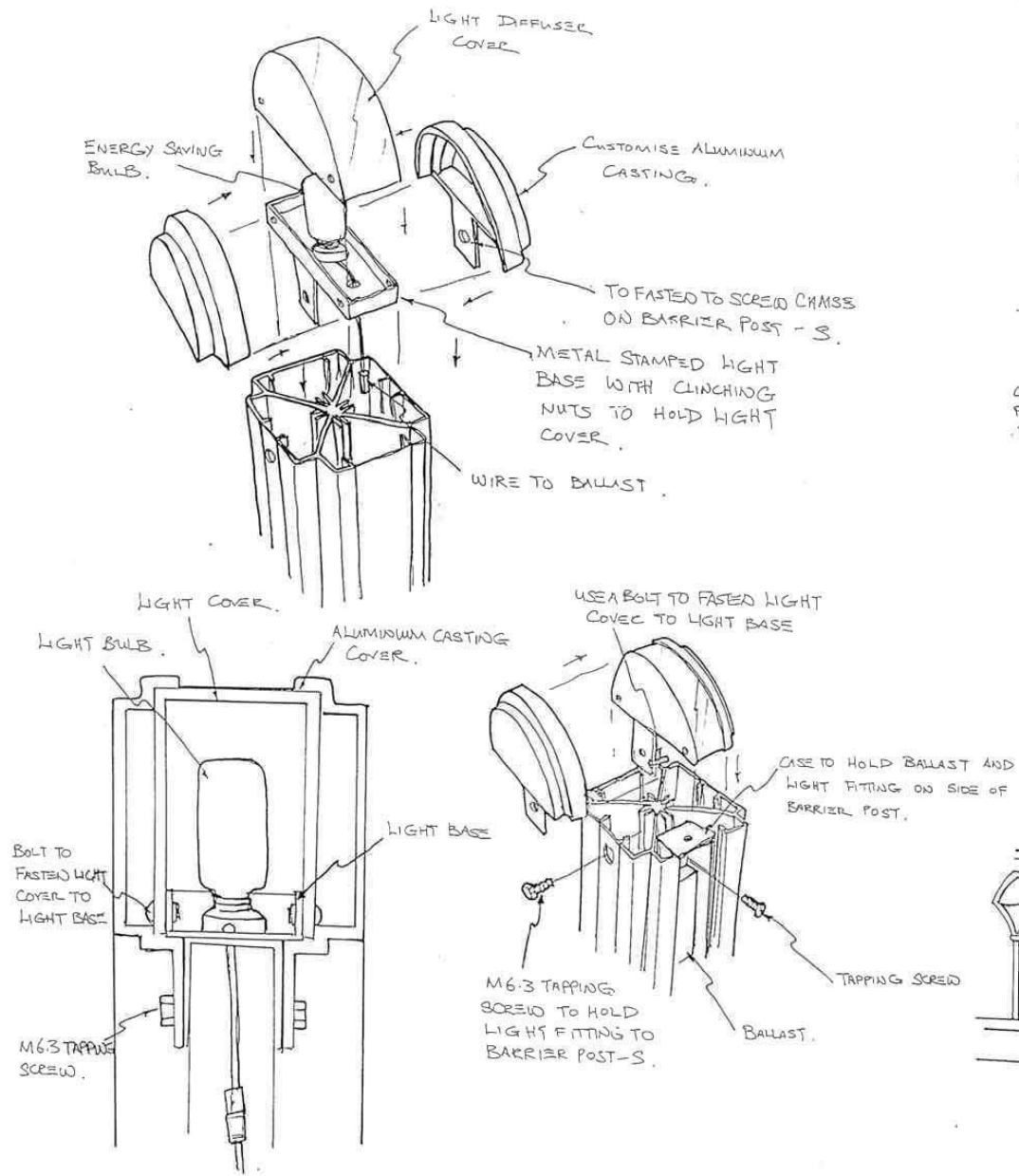


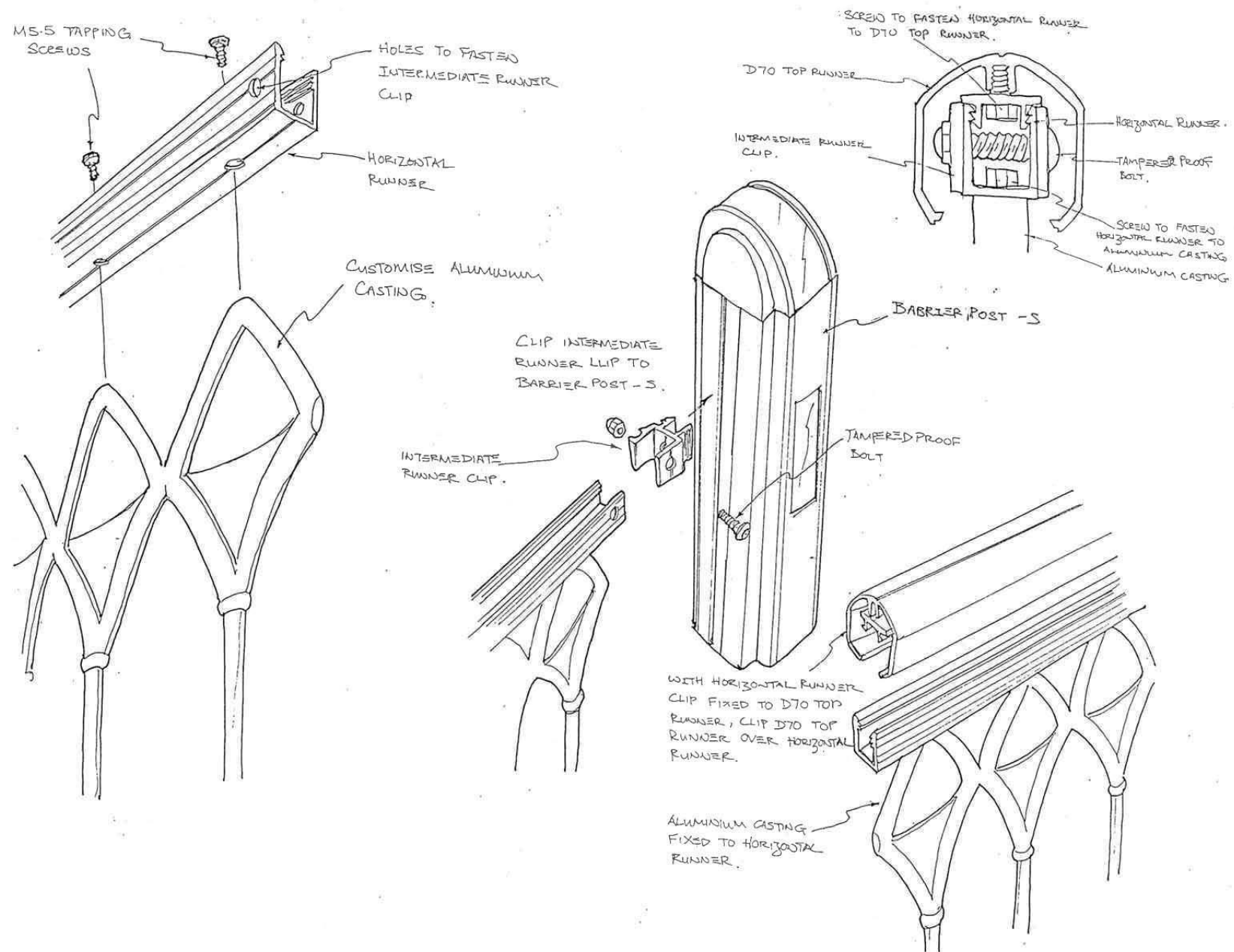
Drawings (Technical Sketching)

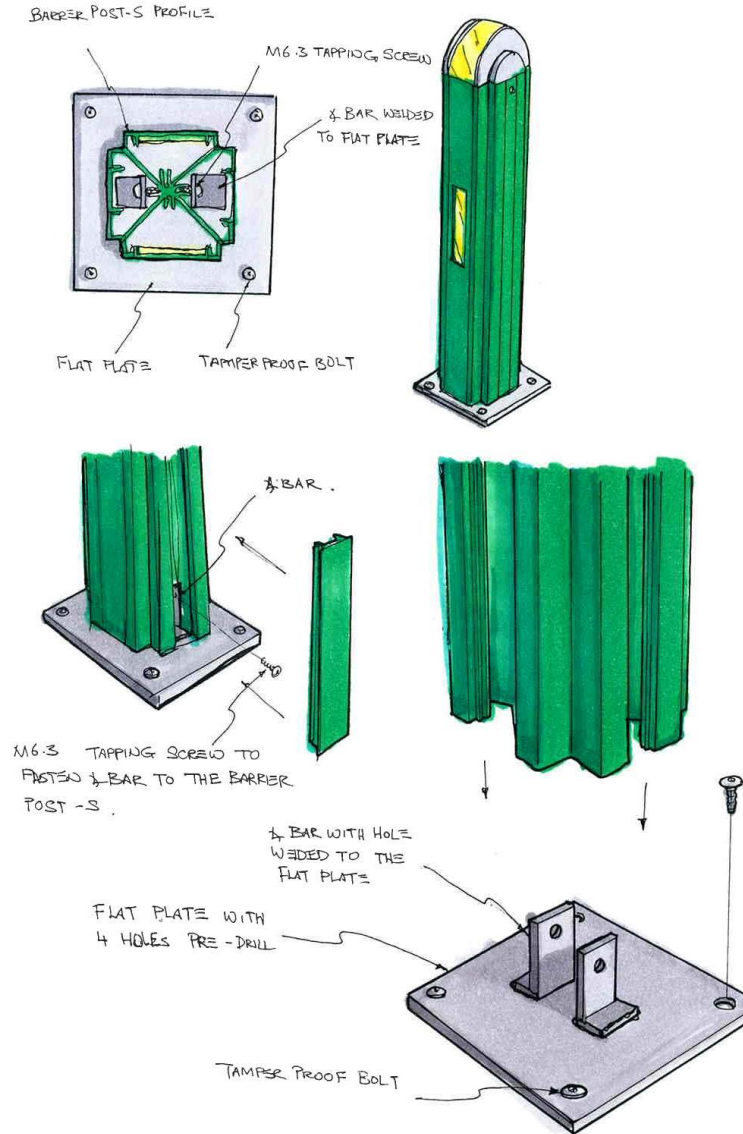
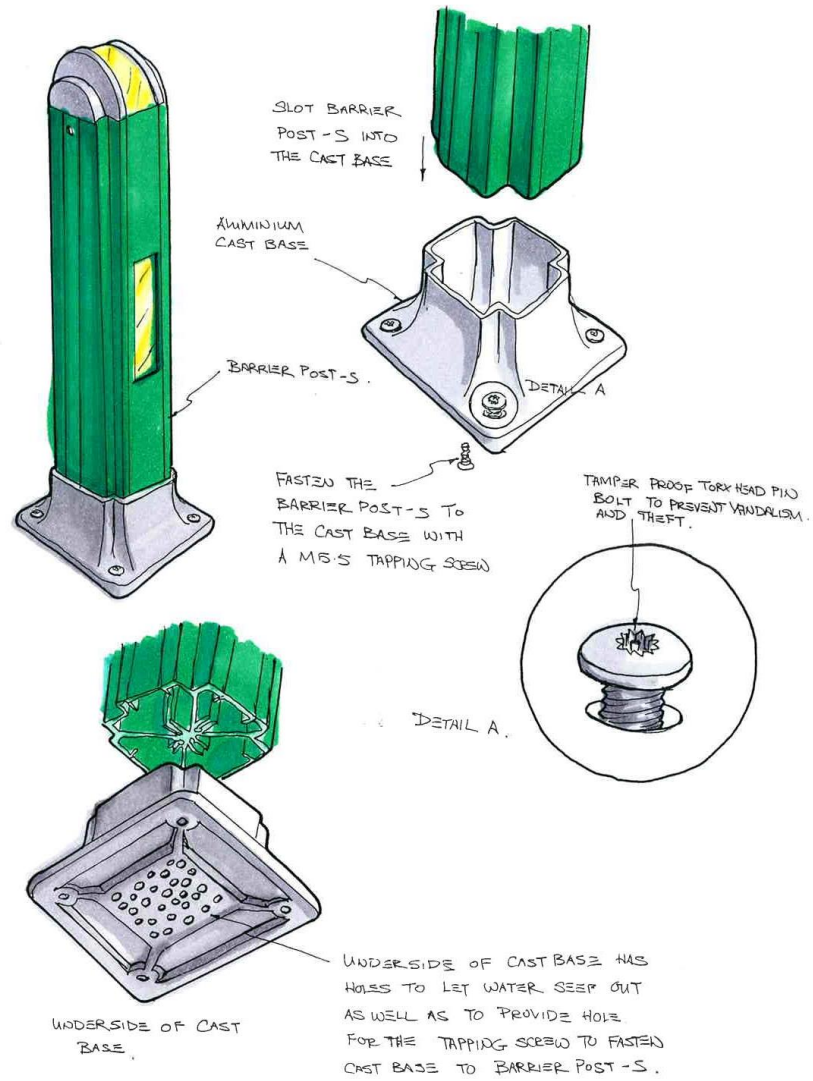
the stage after sketching

more finish image

easier to understand







Getting ready with 5 tips

correct posture

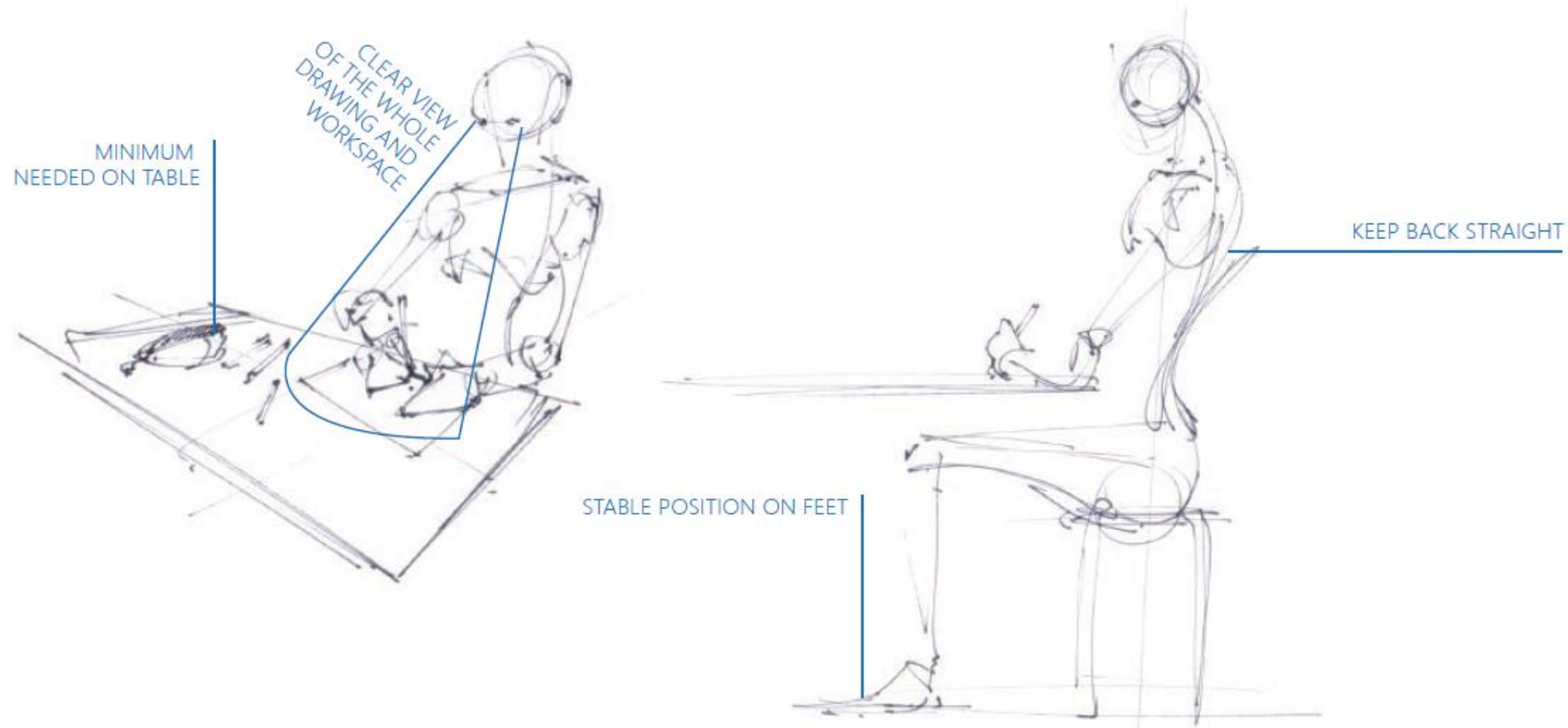
turn your paper

diversify your pen grip

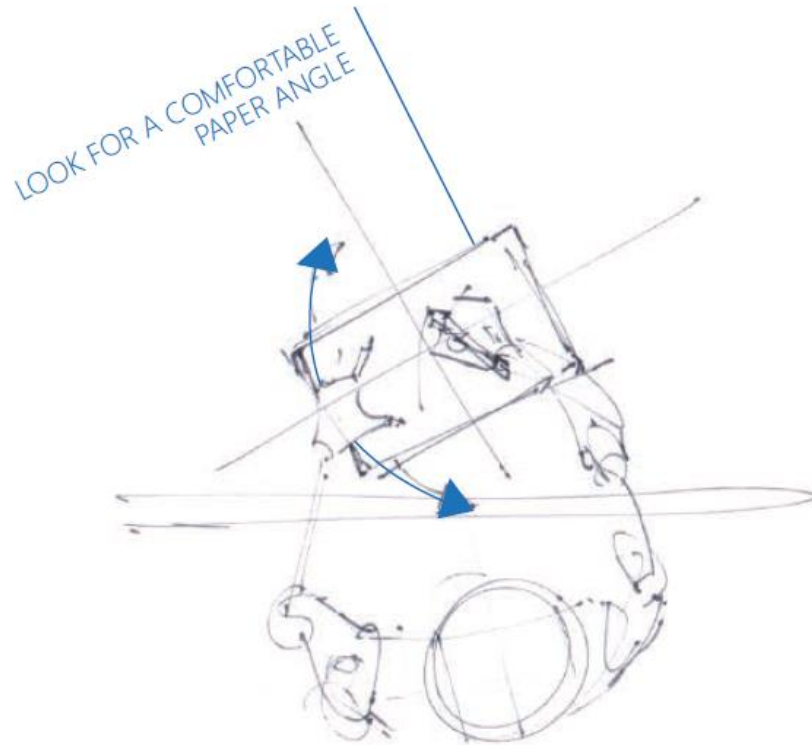
make fluid lines

adopt ghost drawing

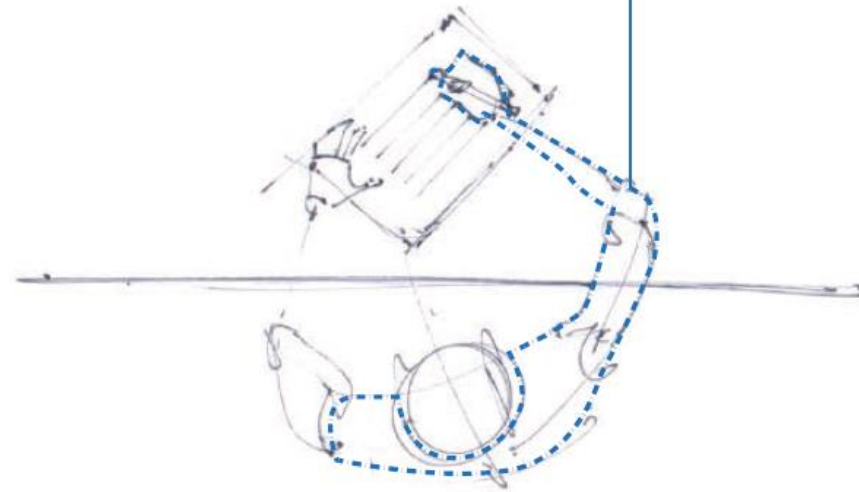
Correct Posture



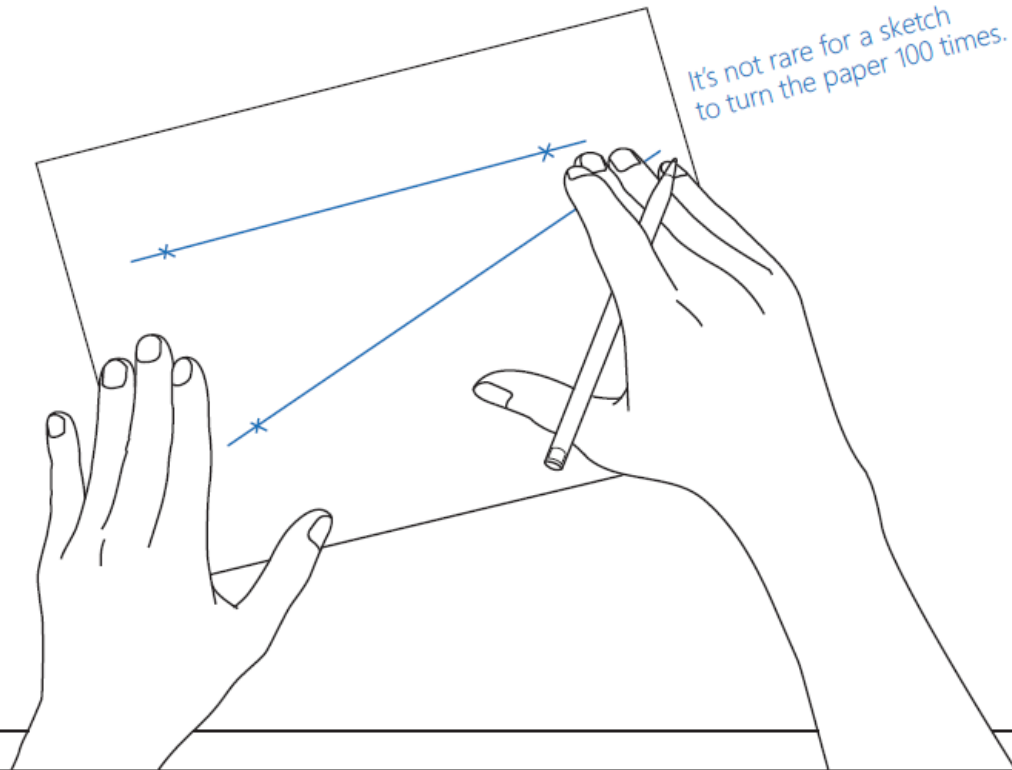
Correct Posture



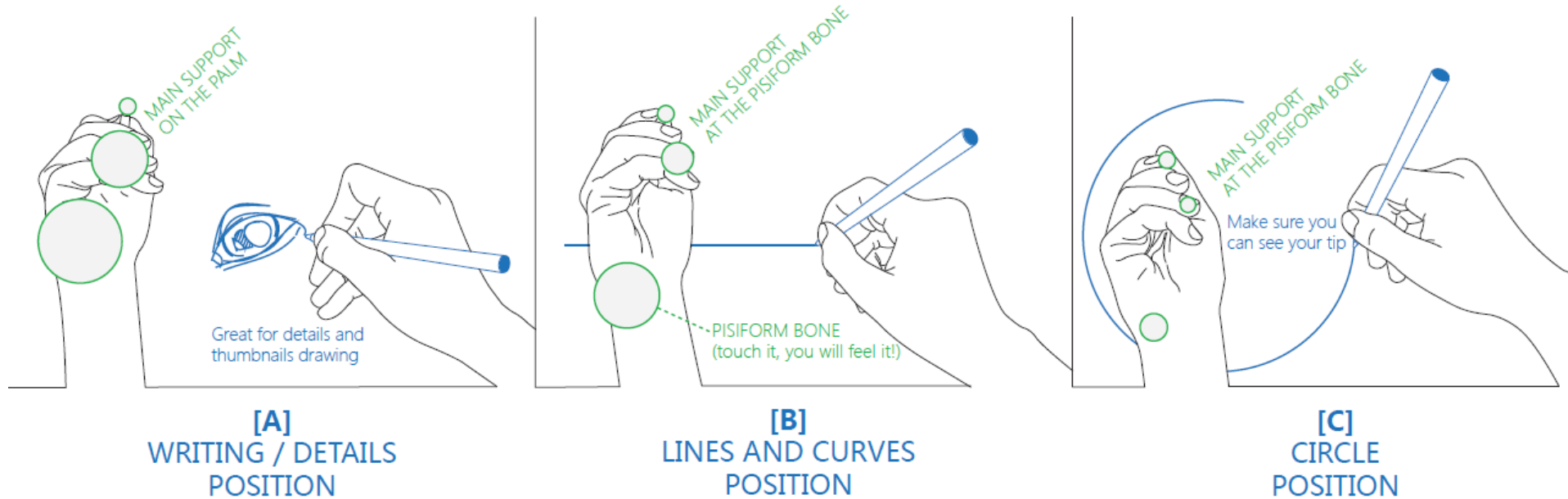
DRAW WITH THE FULL ARM,
ELBOW NOT RESTING ON TABLE
For drawing long lines,
the chest may also move forward a bit.



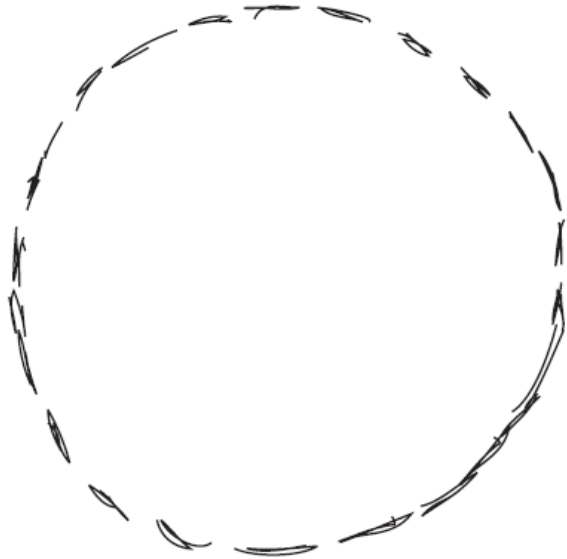
Turn Your Paper



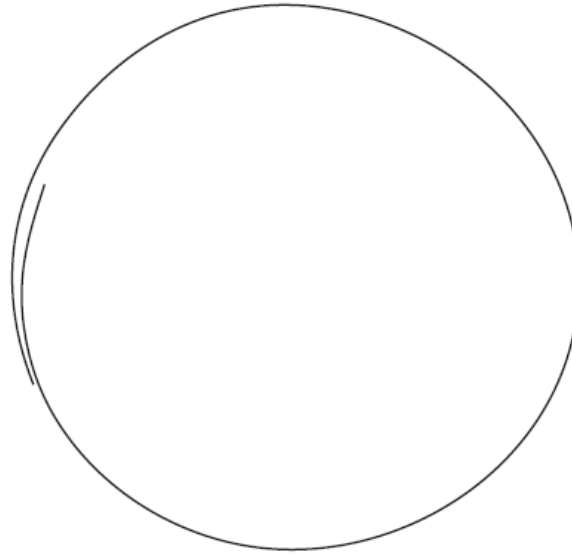
Diversify your pen grip



Make fluid lines

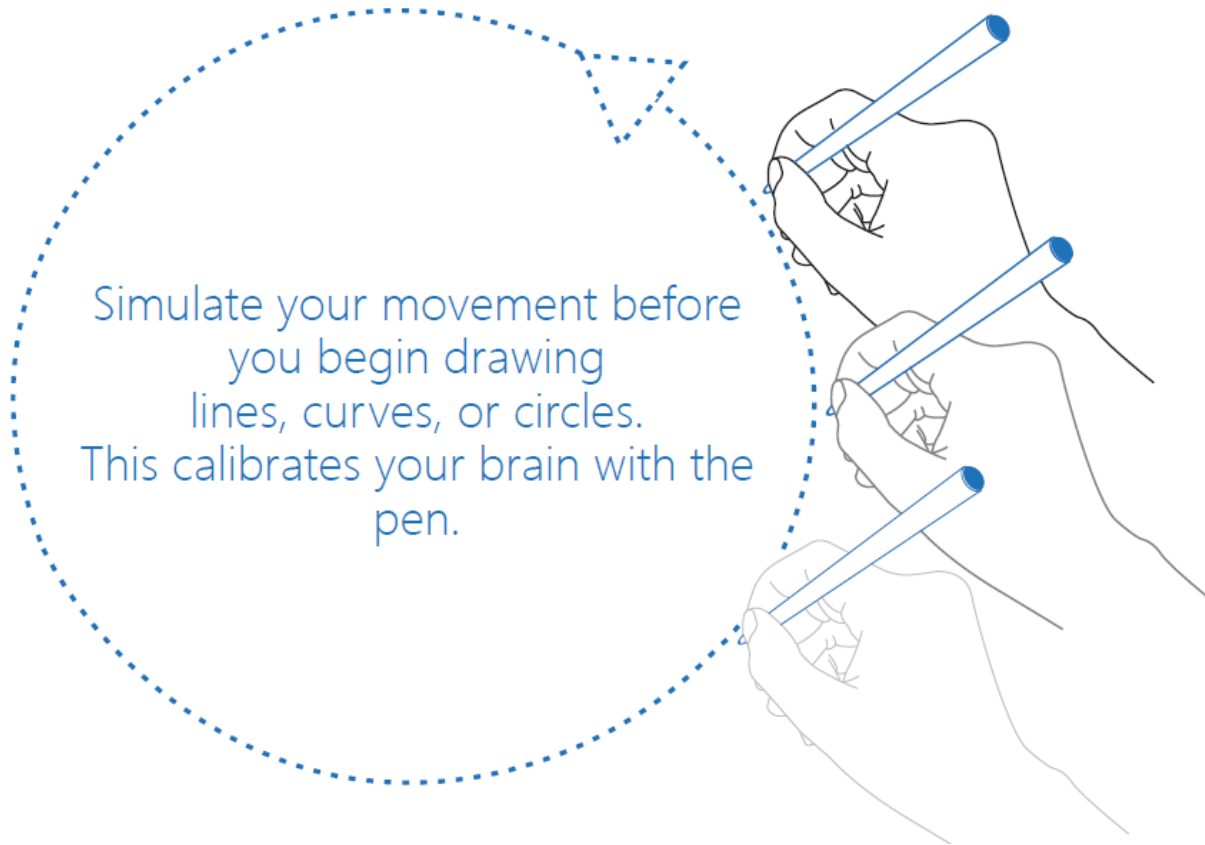


FAILED



PASS!

Adopt the ghost drawing



Mastering the pen

Straight lines



Mastering the pen

PASS!

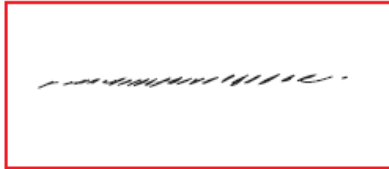


THE ZEN MASTER

You don't have to draw fast; just keep a constant speed.

>> Touch the paper – draw – stop – lift off paper

FAILED



THE FURRY LINE

You're trying to imitate the way artists seem to draw.

>> Work on drawing fluid lines.

If you're afraid of making mistakes, draw with a light wrist.

FAILED

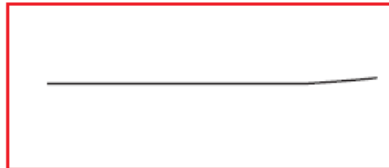


THE CURVED LINE

Either your elbow or your wrist is resting on the table.

>> Draw with your whole arm

FAILED



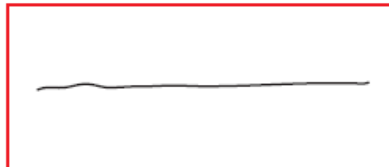
THE TAILED LINE

You didn't end the line with a full stop; instead you drew a line quickly, accelerating at the end.

You did: >> Touch the paper – draw – NOT STOP – lift off the paper

>> Refer to the Zen Master

FAILED



THE SHY LINE

You're trying to do well by going slow, but it's too slow.

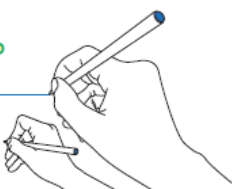
>> Find a minimum speed where you're comfortable but you still get good momentum. You'll get confident with practice.

Mastering the pen

1

RECOMMENDED [B] POSITION

The [A] position is also suitable.

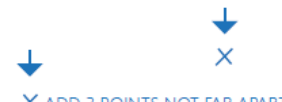


2

↓

↓

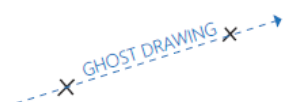
X ADD 2 POINTS NOT FAR APART



3

↓

X GHOST DRAWING X

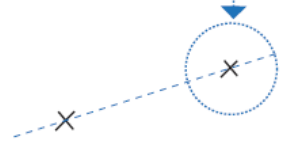


4

STARE AT THE END POINT JUST BEFORE TRACING

↓

X



5

JOIN THE DOTS

YOU CAN STOP YOUR LINE AFTER THE DOT. IT'S OK



6

↓

X ADD A 3rd POINT FURTHER



7

REMINDER

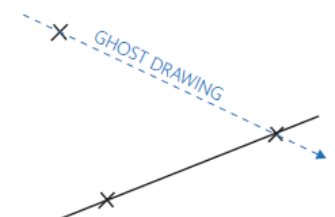
TURN YOUR PAPER WHEN NEEDED



8

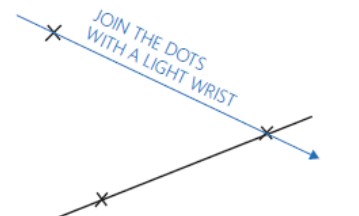
↓

X GHOST DRAWING X



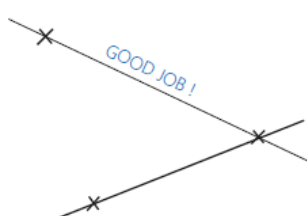
9

JOIN THE DOTS WITH A LIGHT WRIST



10

GOOD JOB !

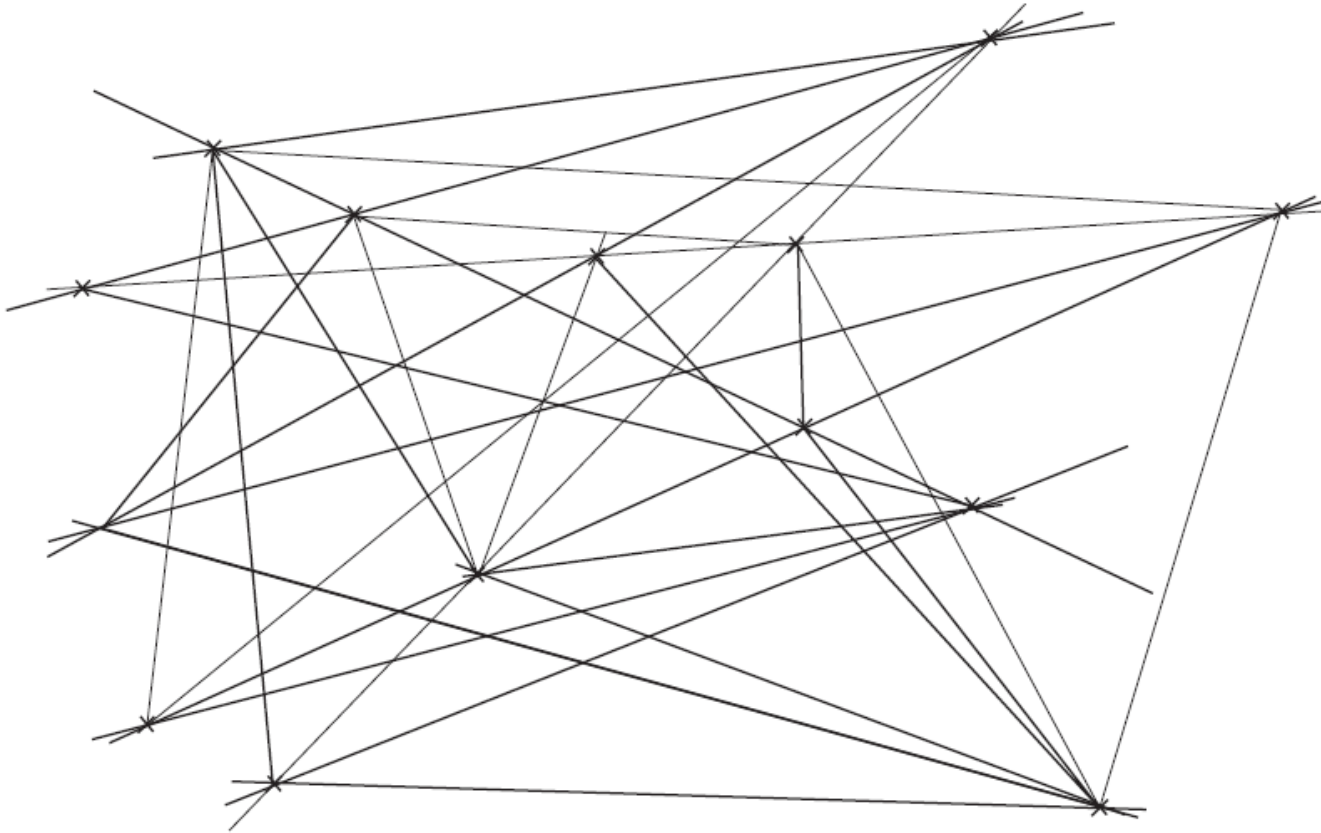


11

REPEAT THE CYCLE x 30

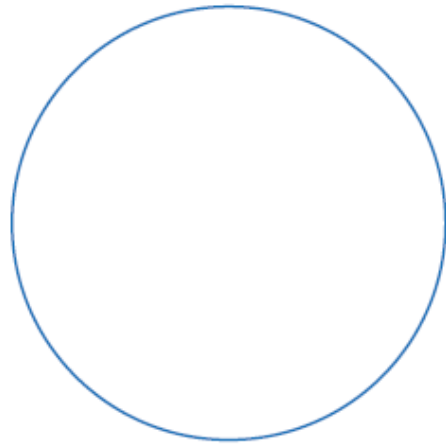
TRAIN TRACING YOUR LINES FROM LEFT TO RIGHT, VICE VERSA

Mastering the pen



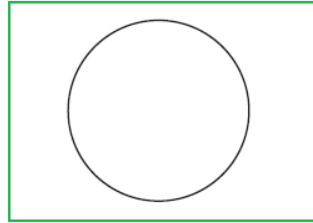
Mastering the pen

Awesome circles

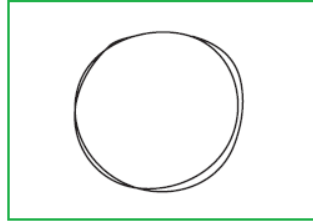


Mastering the pen

PASS!

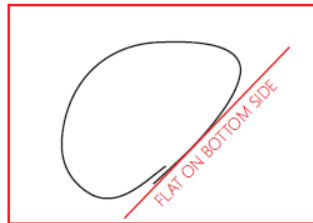


SINGLE RING
Simply awesome!
Master the double or triple ring
before trying this one.



DOUBLE or TRIPLE RING
Recommended
Aim for the momentum to perfect
your circle in the 2nd and 3rd rings.

FAILED

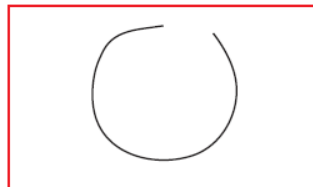


THE POTATO CIRCLE

An elbow or a wrist might be resting on the table; or, your elbow could be knocking against your ribs.

>> Your elbow shouldn't be pressed against your body. Extend your arm and elbow a bit like you're pretending to have chicken wings, and make yourself comfortable!

FAILED

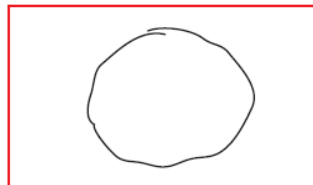


THE UNFINISHED CIRCLE

Too fast!

>> Double or triple the amount of loops you trace around the circle. This will stabilize your movement. When you find your momentum through the ghost drawing, carry that momentum through into the actual drawing.

FAILED

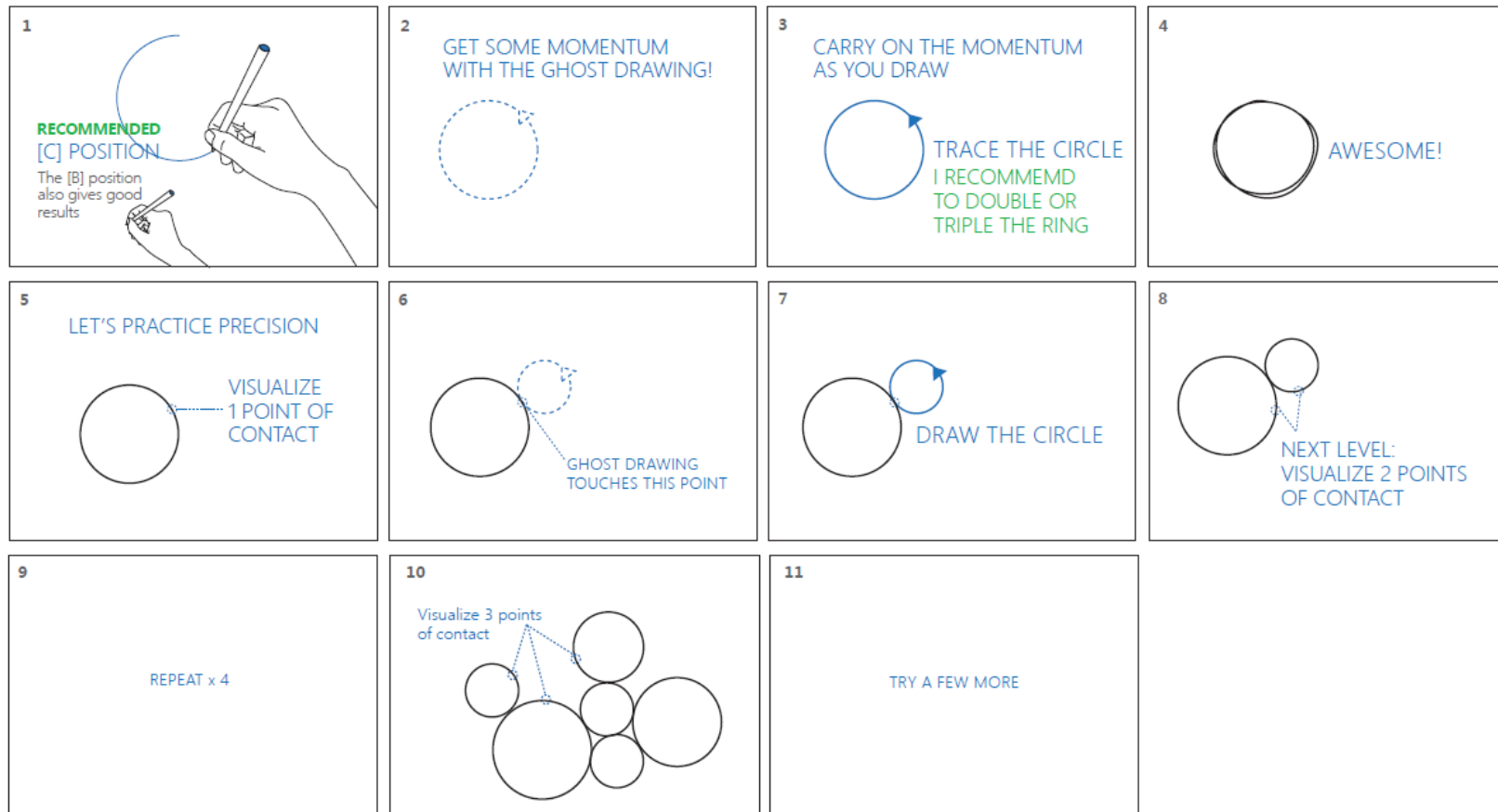


THE SHY CIRCLE

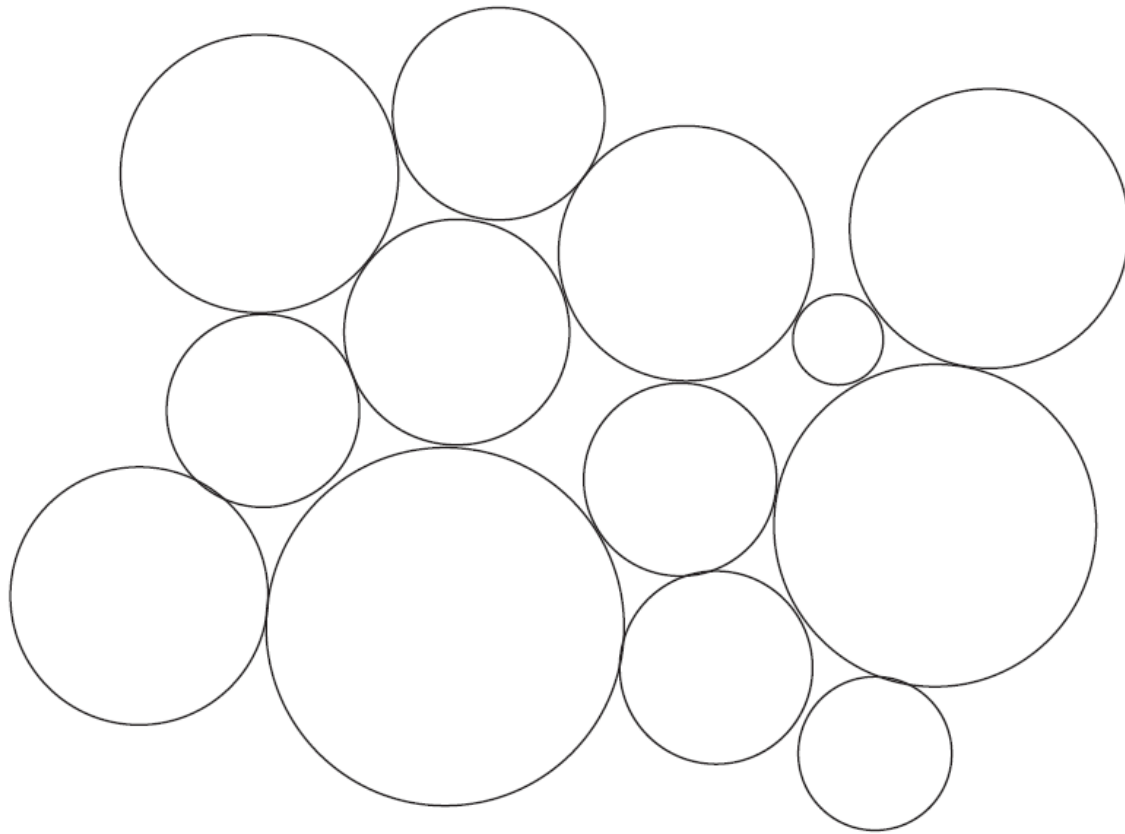
Hesitant and slow; the lines become irregular

>> Focus on the ghost drawing, then make the first two loops of the circle with very light pressure, and the third with a more solid hand.

Mastering the pen

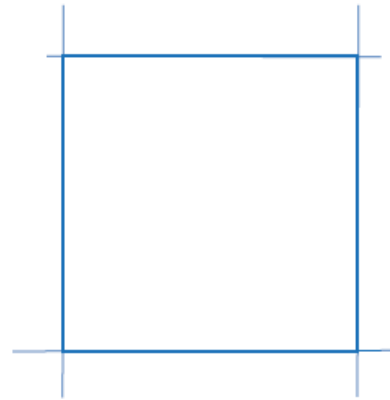


Mastering the pen



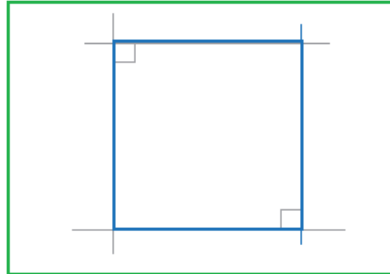
Mastering the pen

The perfect square



Mastering the pen

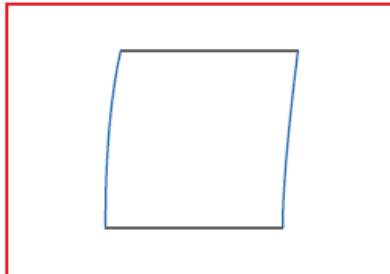
PASS!



THE ARCHITECT

Build your square by sketching light lines first. Then bold the square.

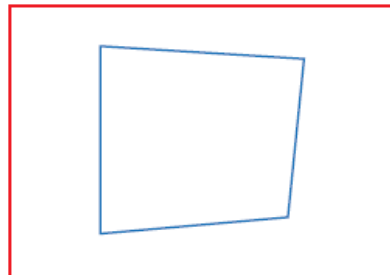
FAILED



THE CURVY CASE

You might be drawing with your elbow or wrist on the table
>> Review how to master the straight line

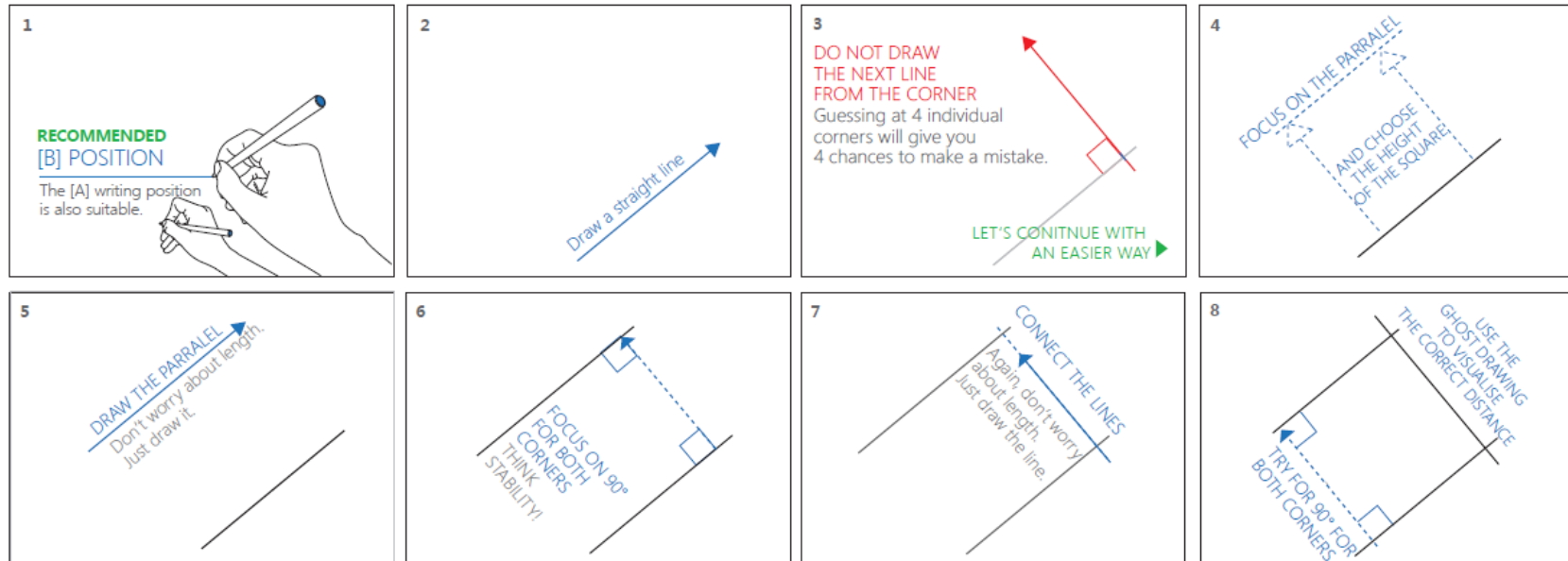
FAILED



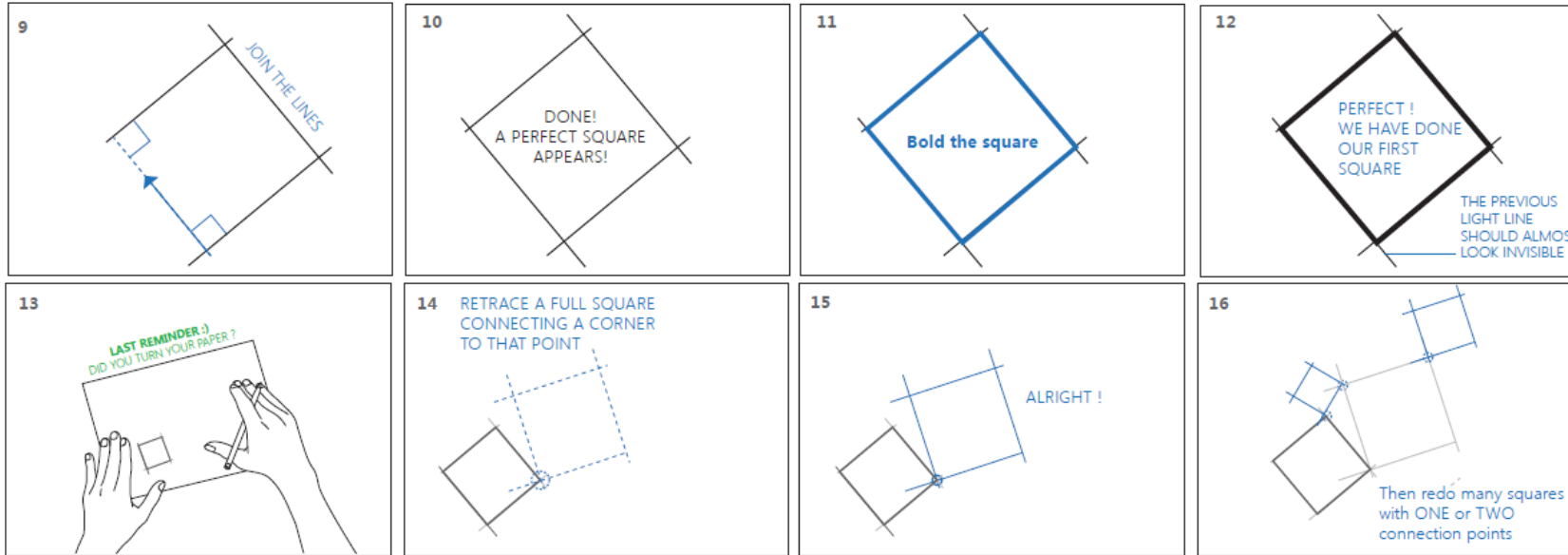
NON-PARALLEL LINES

You felt like you were doing well when you traced the lines, but it still comes out awkward.
>> Sometimes your vision's been skewed, especially if there are already other lines on the paper. Relax, focus, and take your time.
>> As you turn the paper, keep visualizing the next line. Do not look elsewhere. Doing so may disconnect your brain from the shape. Keep focus, and your speed and precision will benefit.
>> If your line is parallel to your paper edge, use it as reference.

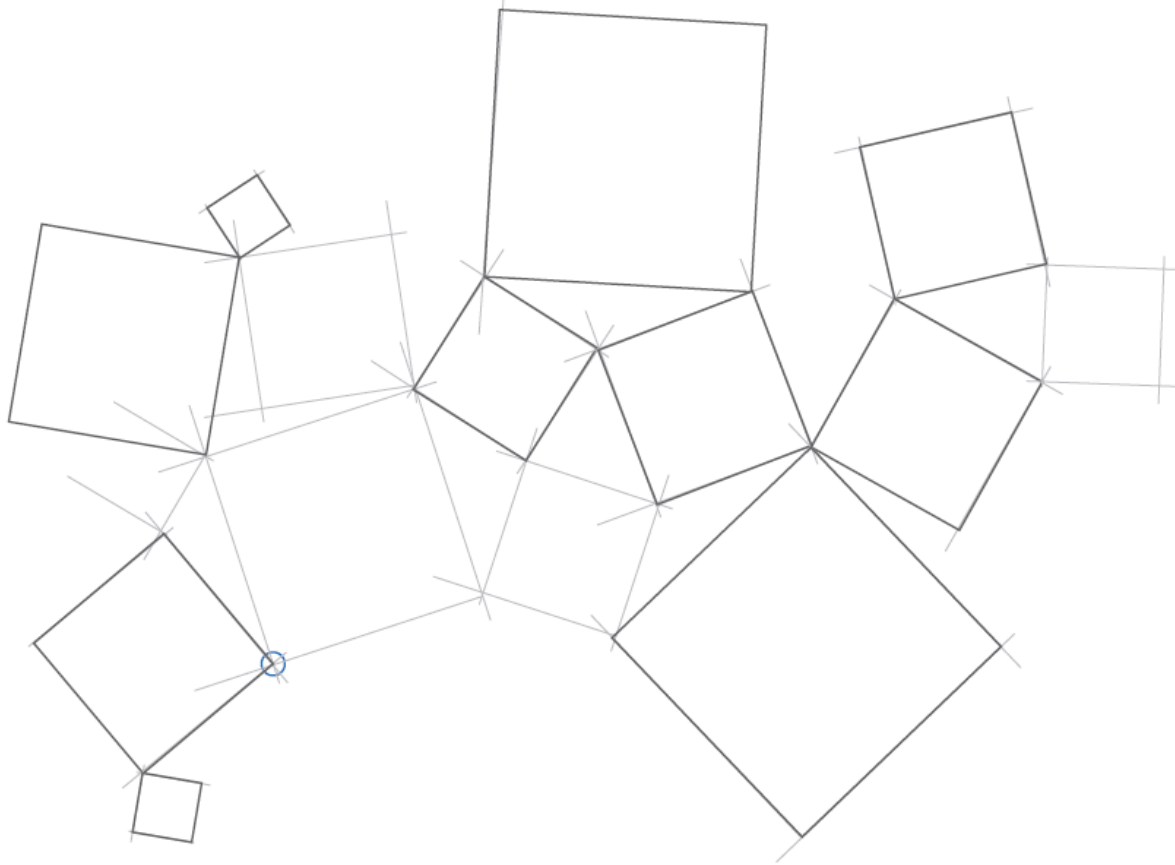
Mastering the pen



Mastering the pen



Mastering the pen



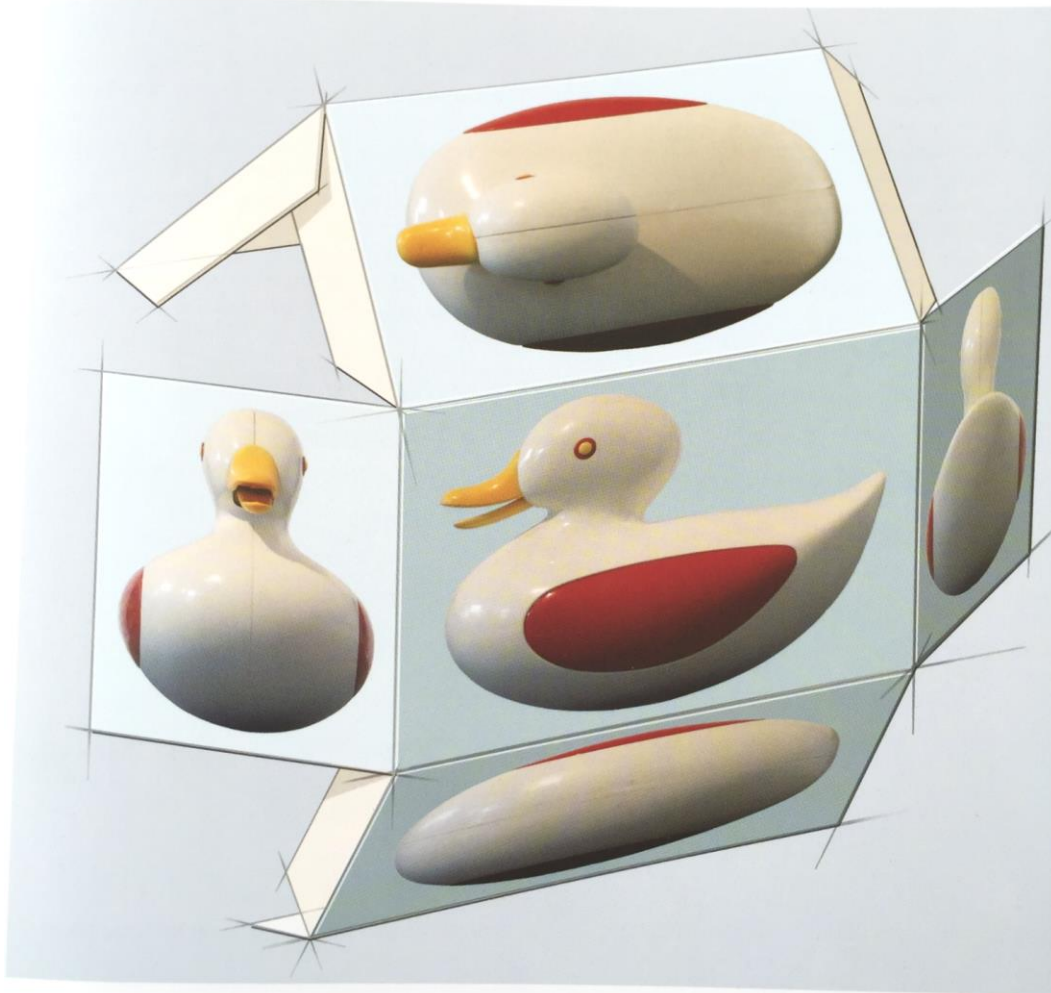
5 Essentials to Good Technical Sketching

frontal sketch

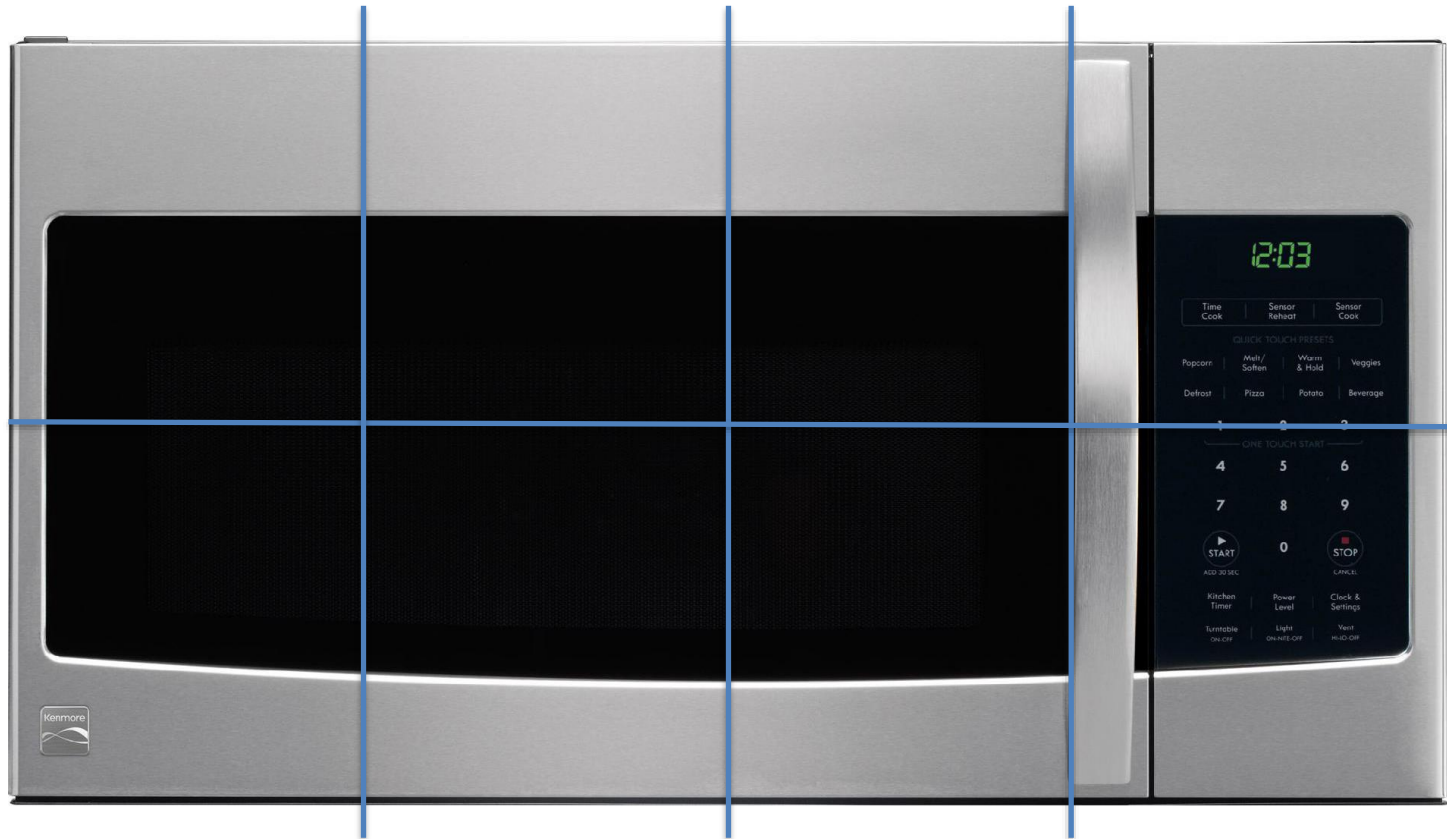
to scale

clean lines

Front view sketching exercise



Front view sketching exercise



5 Essentials to Good Technical Sketching

frontal sketch

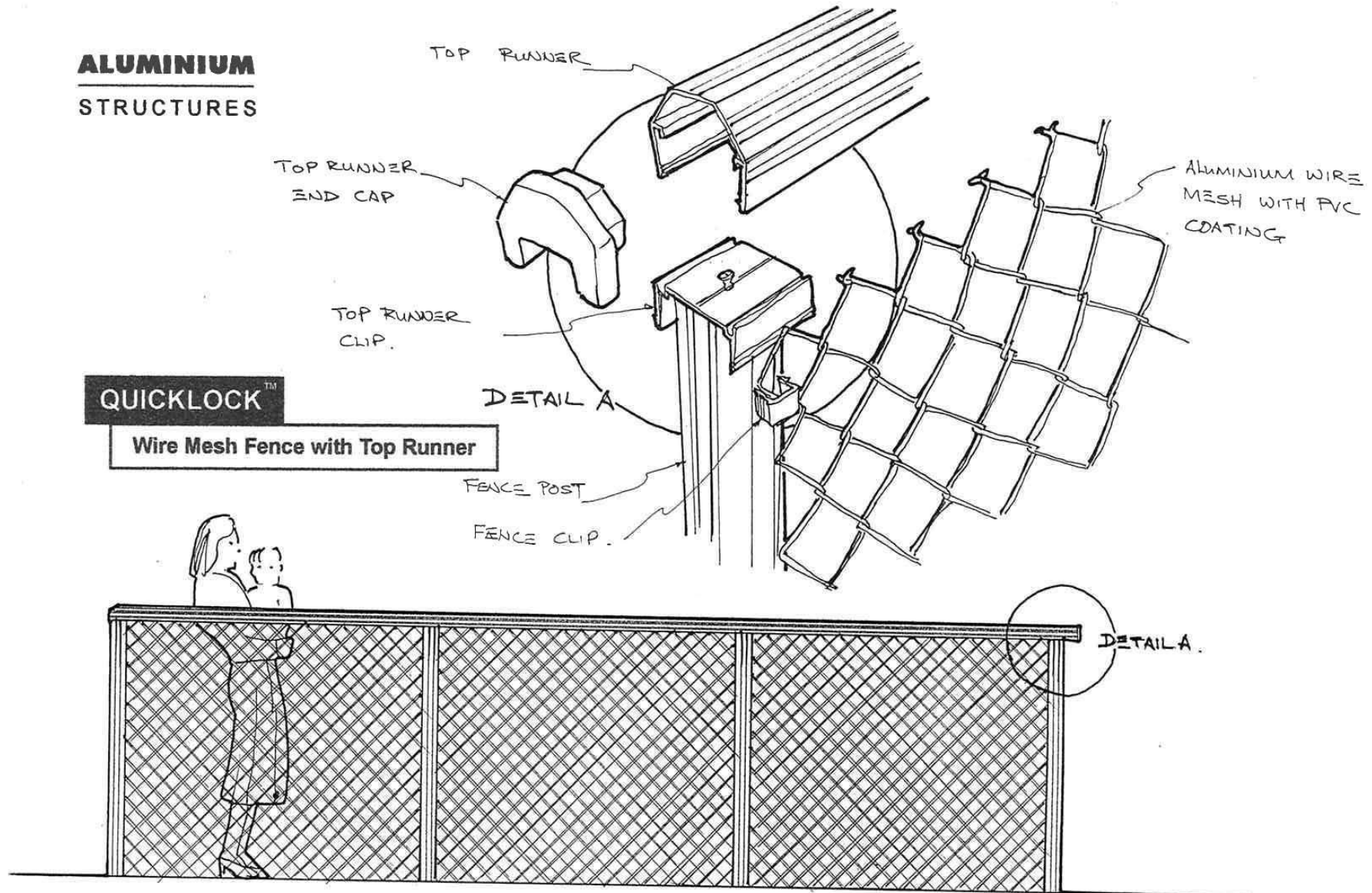
to scale

clean lines

line weight

shading

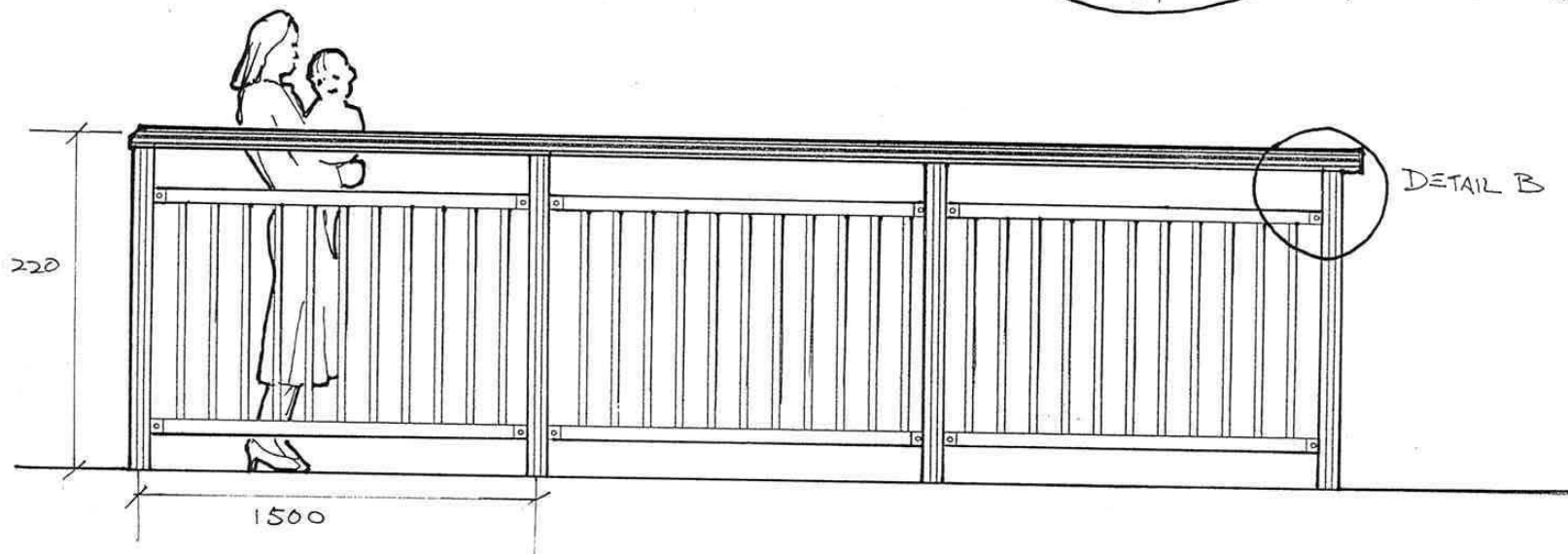
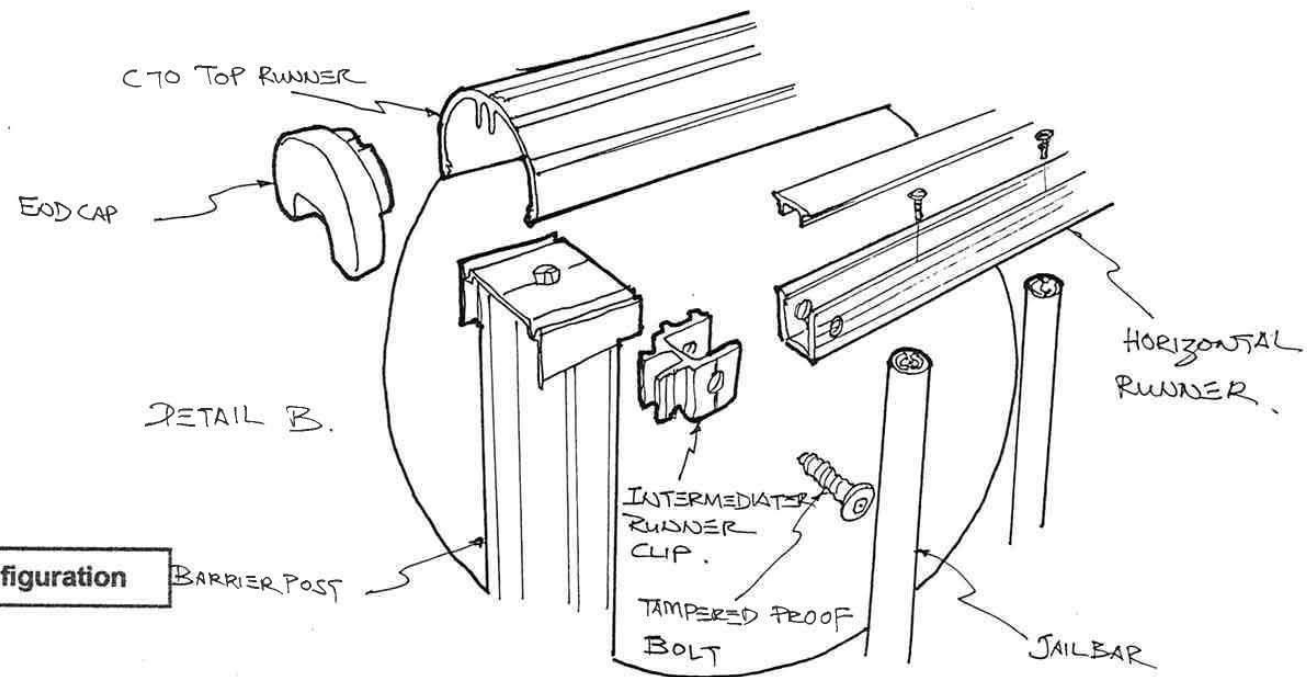
ALUMINIUM STRUCTURES



ALUMINIUM STRUCTURES

QUICKLOCK™

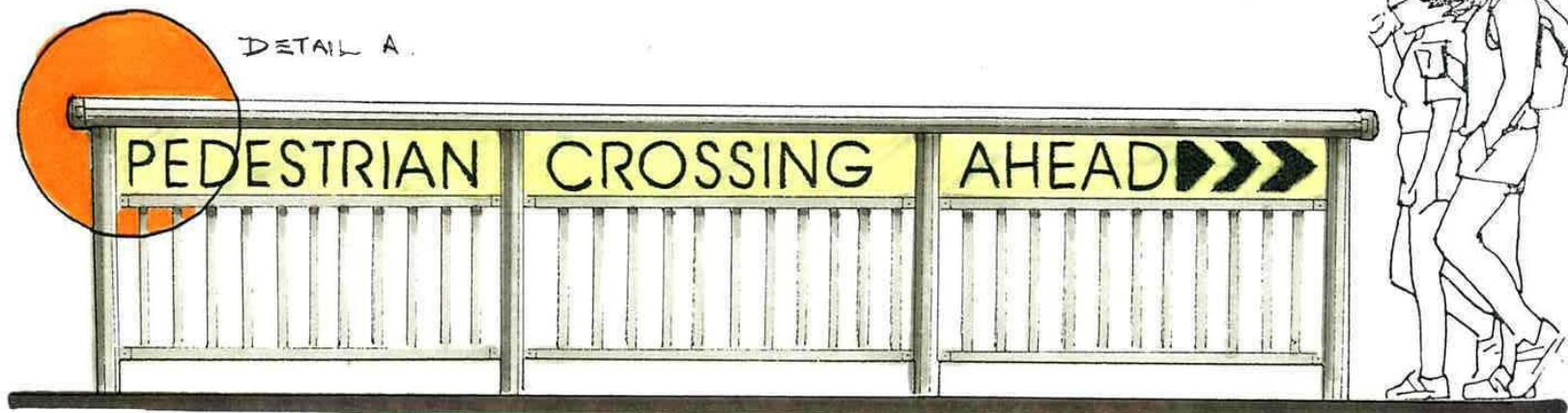
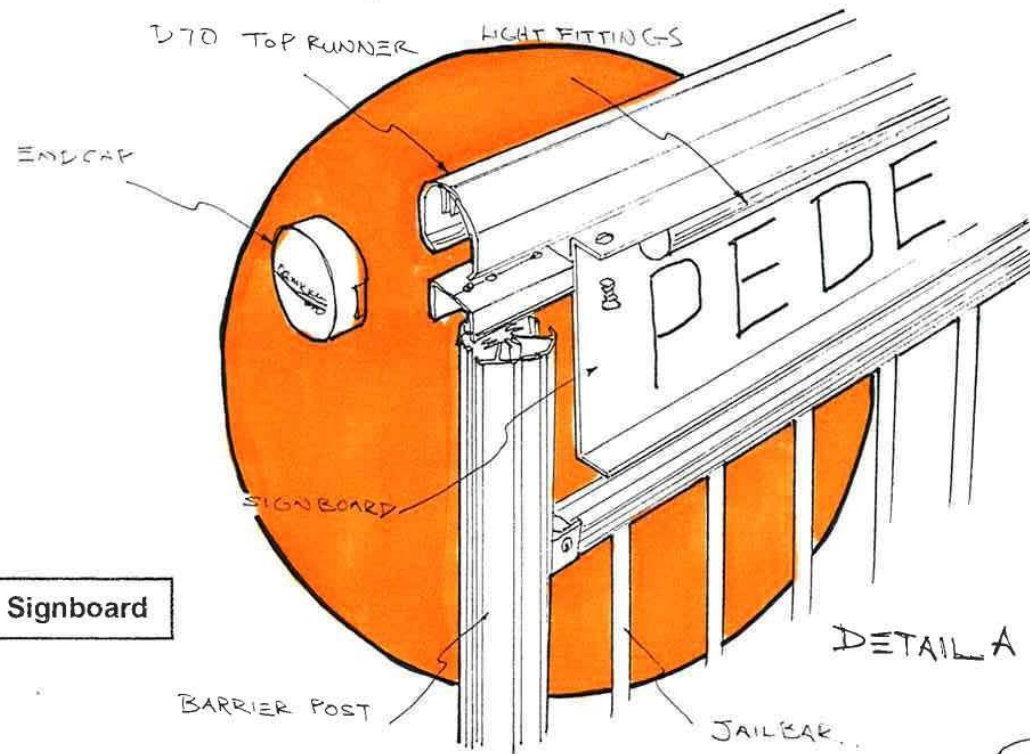
C70 Jailbar Barrier Configuration



ALUMINIUM STRUCTURES

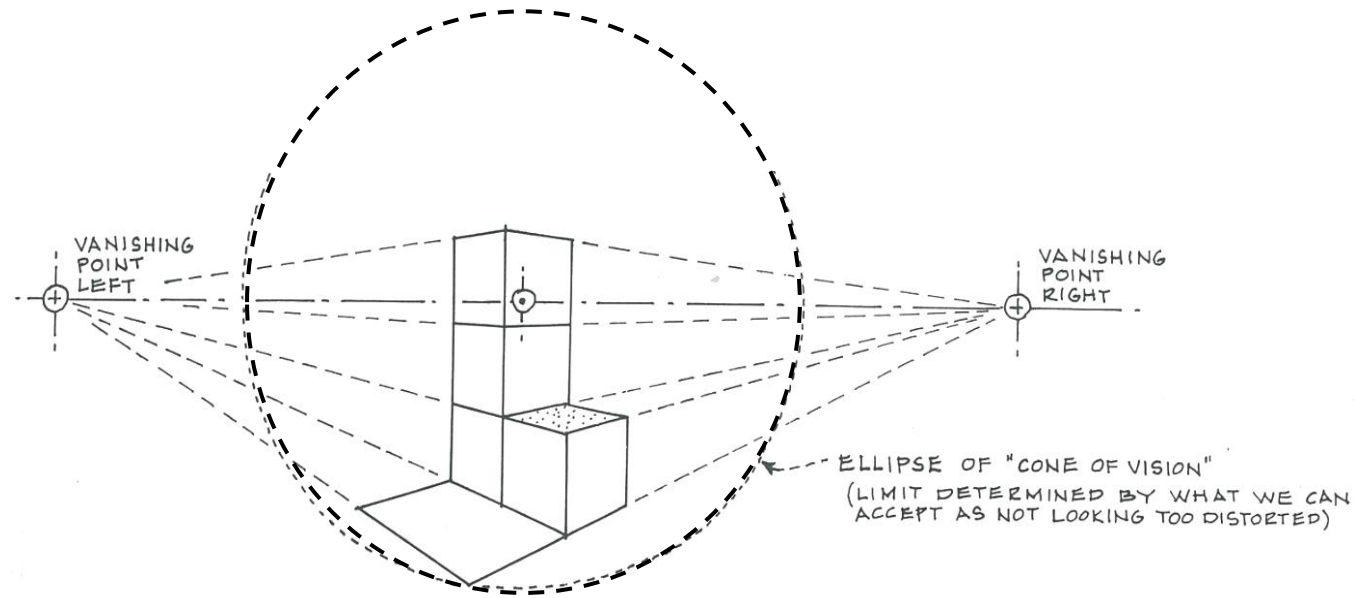
QUICKLOCK™

D70 Jailbar Barrier with Signboard

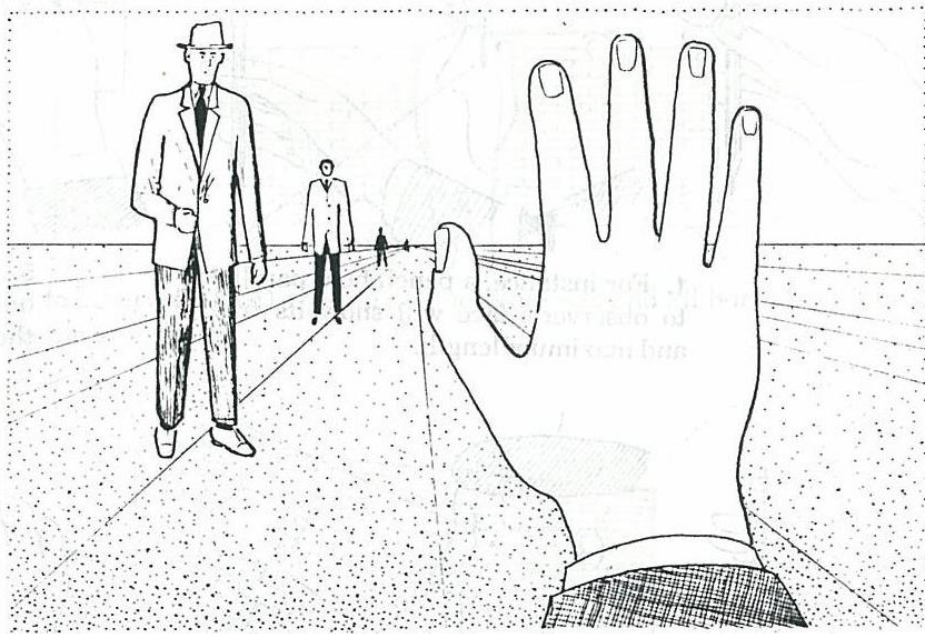


Perspective drawing

parallel lines appear to converge to a “vanishing point”



Objects appear smaller as their distance from the observer increase







|| ONE-POINT PERSPECTIVE

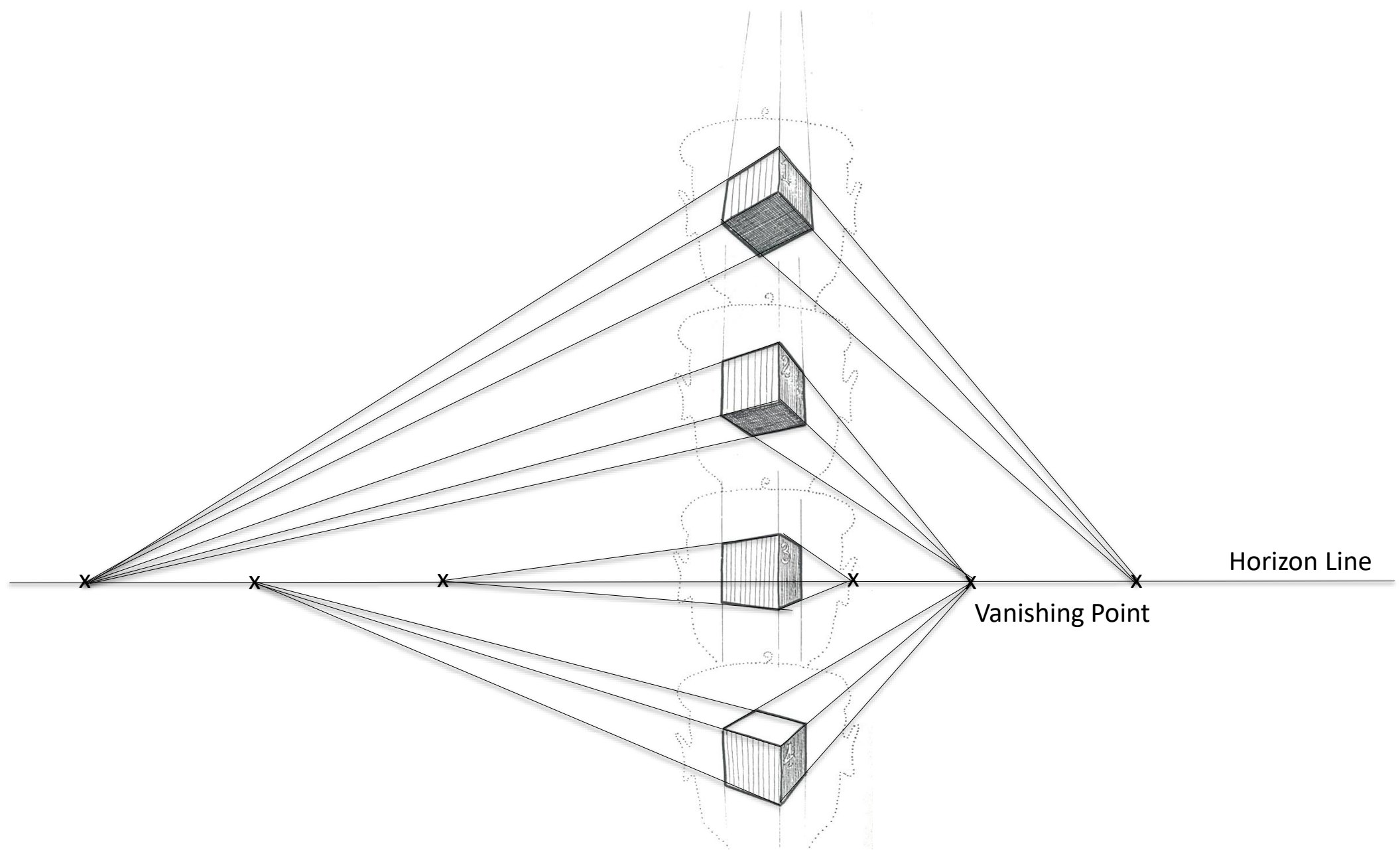


TWO-POINT PERSPECTIVE
APPROX. 45° SET-UP

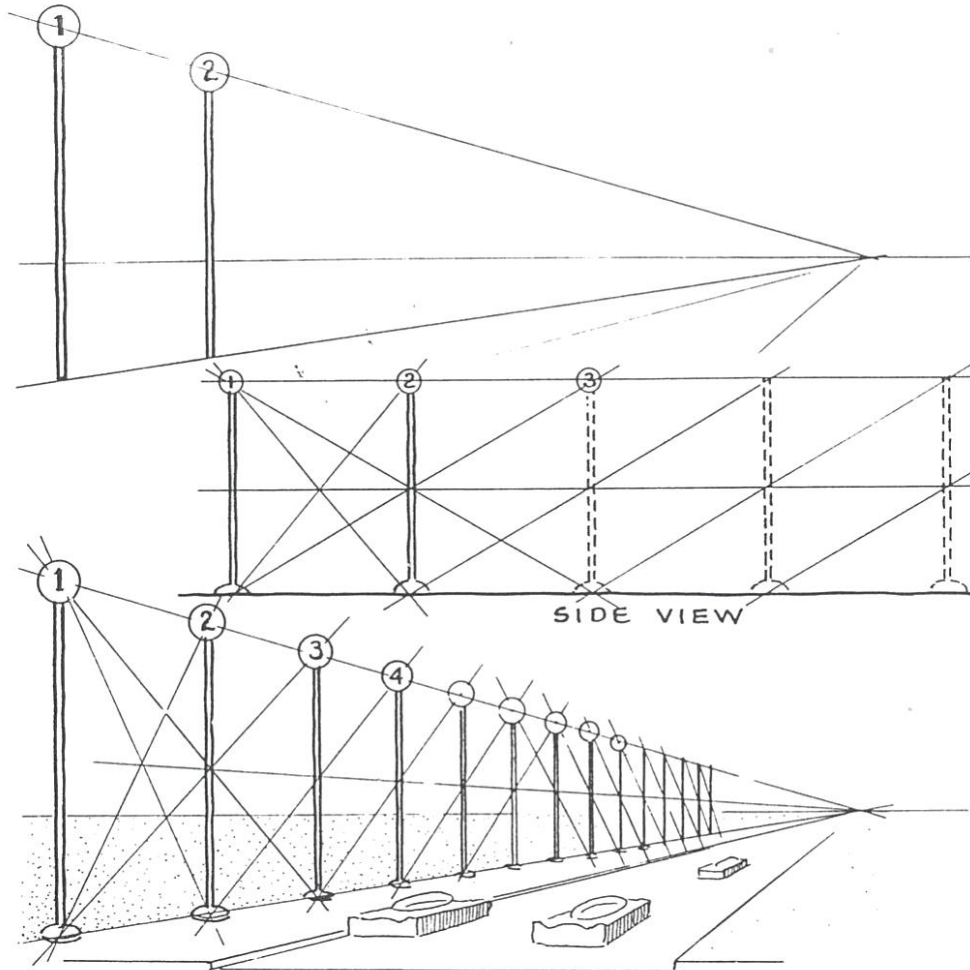


THREE-POINT
PERSPECTIVE
AERIAL

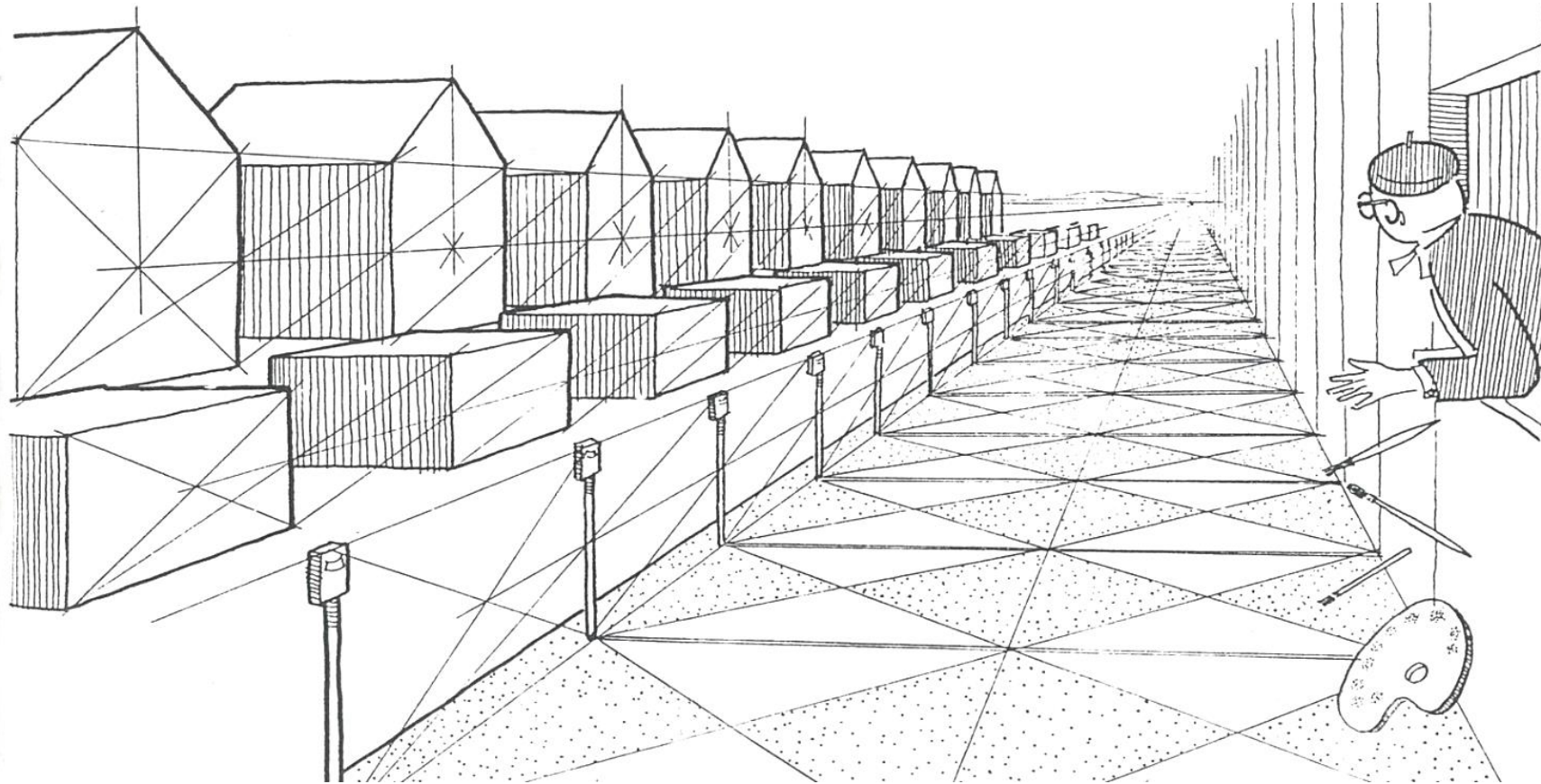
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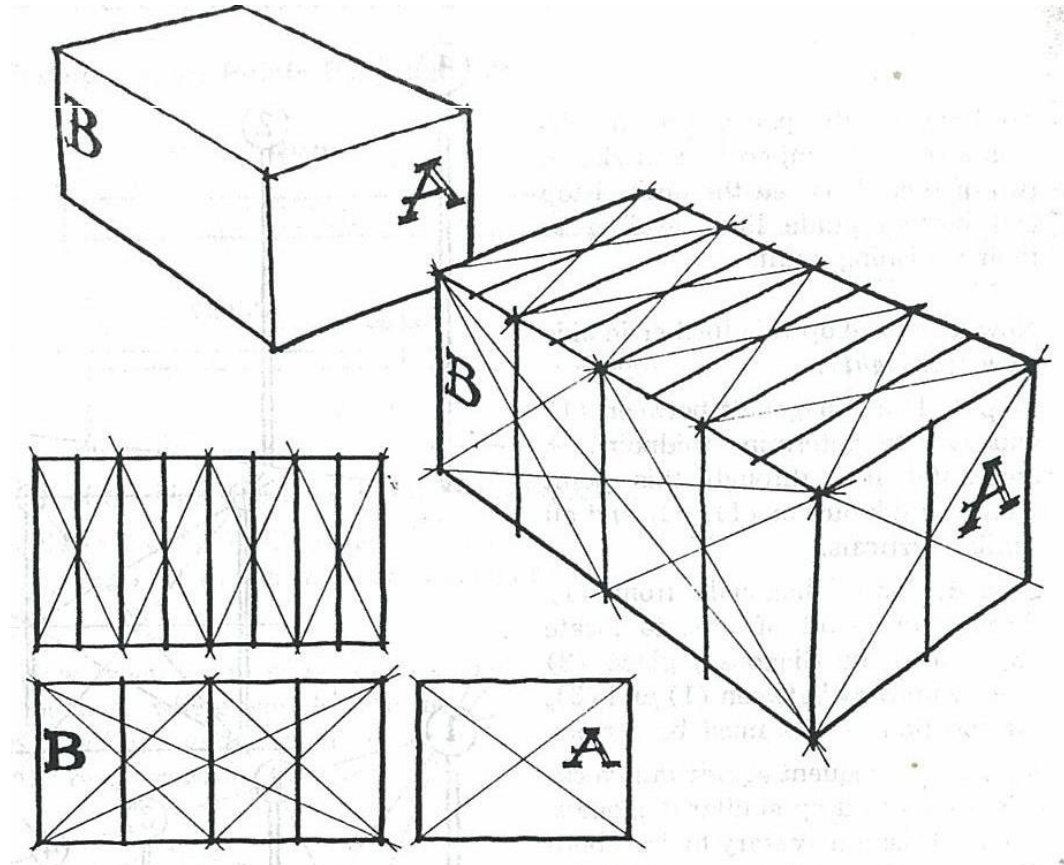
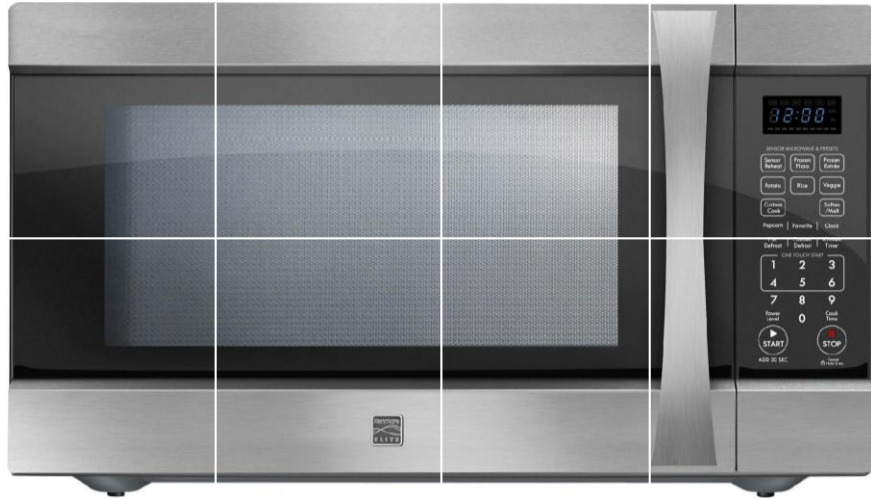


Equally-spaced receding elements

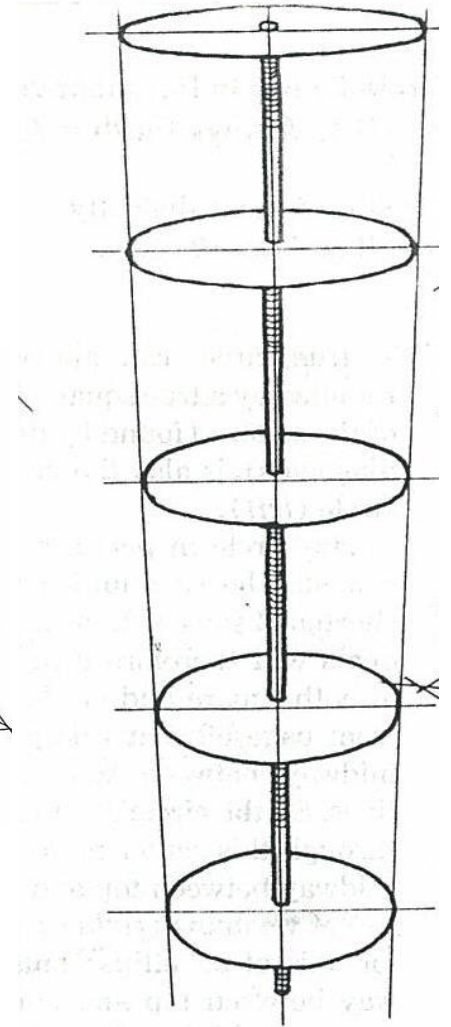
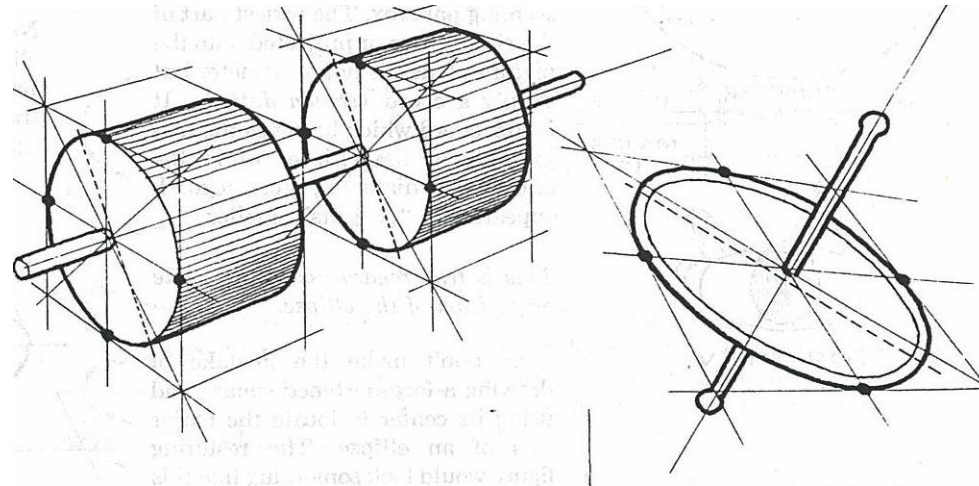
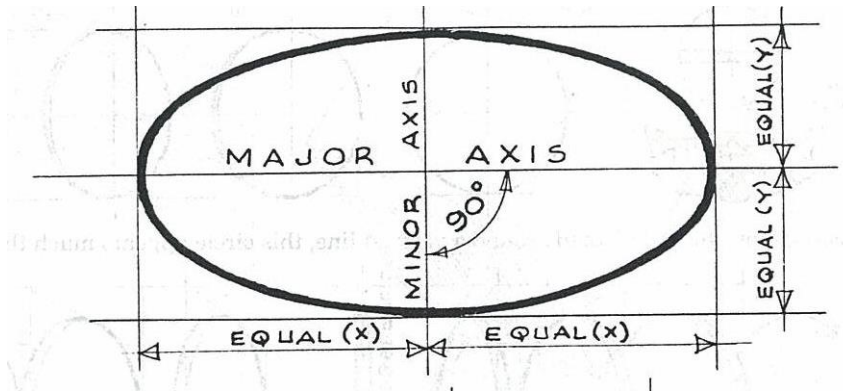


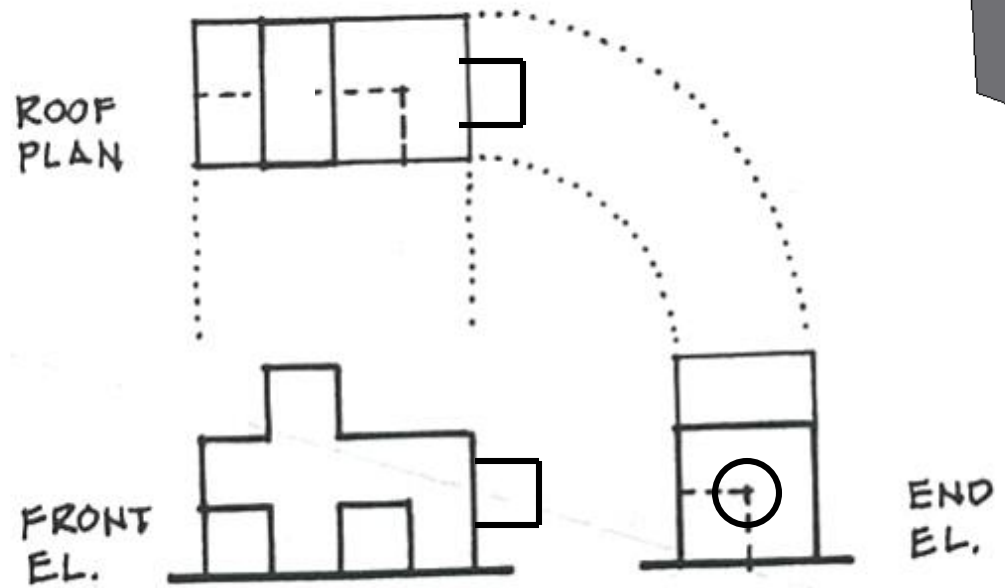
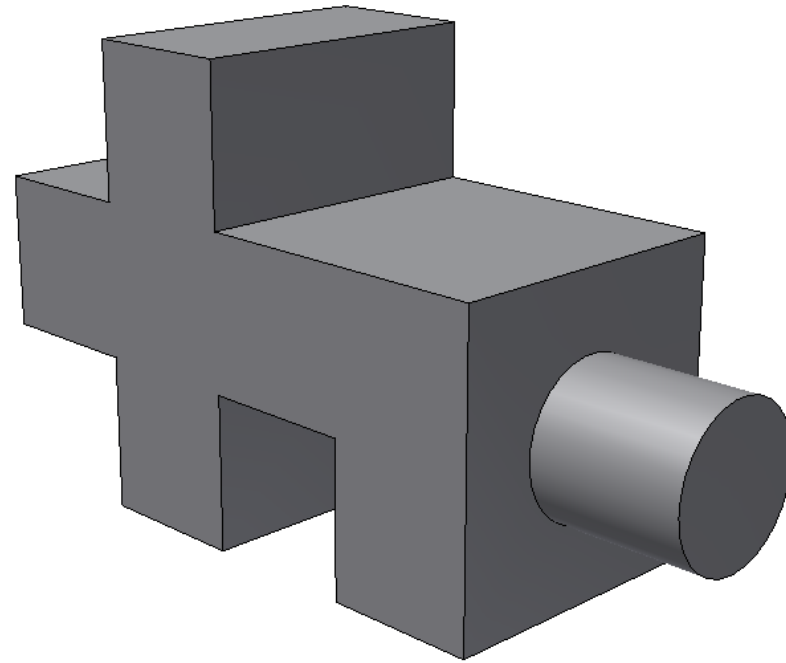
Equally-spaced receding elements





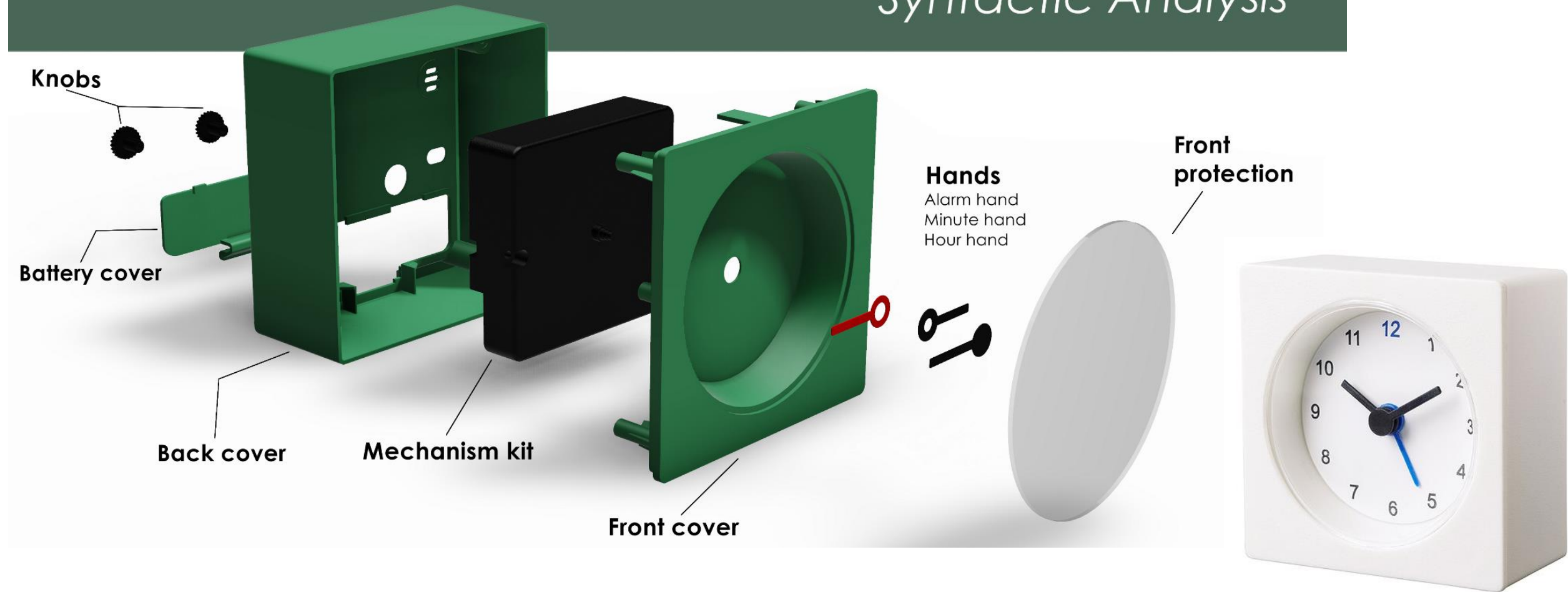
Drawing cylindrical part





IKEA Alarm Clock

Syntactic Analysis



Reading List

- Eissen, K., Roselien, S. (2019). *Sketching: Drawing Techniques for Product Designers*. Laurence King Publishing.

End